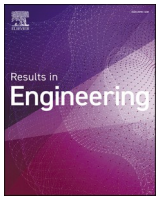
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Review article



Risk-based asset integrity management in the oil and gas industry from traditional to machine learning approaches: A systematic review

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ABSTRACT

Oil and gas operations are categorized as high-risk because they involve numerous equipment and complex processes. Asset integrity management (AIM) aims to mitigate the risk of failure resulting from degradation with corrosion as the primary cause and to maintain equipment safety and functionality. The risk-based inspection (RBI) methodology is one of the AIM processes that considers risks in decision-making to prioritize inspection and maintenance. This paper provides a comprehensive review of risk-based studies in the context of AIM activities. Risk-based AIM is categorized and reviewed based on risk analysis methods, including quantitative, qualitative, semi-quantitative, probabilistic, deterministic, hybrid probabilistic-deterministic, and dynamic or traditional risk. Most research areas used in case studies focus on pipeline applications. Analysis tools for risk assessment and control applied in risk-based AIM, including the evolution of tools from traditional to machine learning approaches, are examined. The current trends and future research opportunities for applying risk-based AIM are also discussed. This study offers risk assessment models for researchers and oil and gas industry practitioners that fit their specific requirements.

1. Introduction

The oil and gas sector remains the globe's primary energy source, involving many complex equipment and processes, so it is categorized as a high-risk industry. According to Rachman et al. [1], the oil and gas production and processing facilities face several degradation mechanisms, potentially leading to failures such as leaks, ruptures, or bursts, which can profoundly affect an organization's safety, finances, and environment. Pipeline systems (pipelines, valves, pumps, and other essential equipment), which involve production equipment in offshore platforms, are critical facilities for offshore oil and gas production due to limited space, where accidents could quickly lead to catastrophic consequences [2]. Floating production storage and offloading (FPSO) systems are exposed to severe environmental loads due to extreme sea conditions, including large waves and strong currents. These conditions increase the likelihood of accidents, such as oil and gas leaks. They lead to the dispersion of hydrocarbons and the formation of flammable clouds, resulting in potential fire and explosion risk [3].

Asset integrity management (AIM) is carried out to prevent and mitigate the risk of equipment failure resulting from degradation, minimize associated risks, and ensure continued safe operation. Equipment degradation occurs through corrosion as the primary cause of pipeline failure [4]. The AIM program utilizes risk management for sustainable and continuous self-assessment and improvement [5]. Ratnayake et al. [6] reported that AIM ensures the personnel, systems, processes, and resources required to maintain integrity are appropriately established, effectively utilized, and suitable for their purpose throughout the asset life cycle. This life cycle encompasses the design, construction, installation, commissioning, and operation phases. AIM is focused on enhancing operational discipline and minimizing incidents by implementing an excellent asset integrity program to address the root causes of incidents. Ciaraldi [7] described ten reasons why asset integrity implementation often fails; one of the reasons is failure to prioritize what needs to be done effectively; thus, an essential element of implementing a successful asset integrity strategy is ensuring that the right work is performed in the right place at the right time.

The risk-based inspection (RBI) is a systematic approach for decision-making that considers risk to optimize the prioritization of inspections and maintenance activities. The risk of failure (RoF) is calculated as a result of integrating the probability of failure (PoF) and the consequence

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of failure (CoF). RBI is an element of AIM that focuses on maintaining the capability of physical facilities to operate their intended functions effectively and efficiently while ensuring the safety of personnel and protecting the environment [8]. Deciding which equipment should be inspected, selecting the appropriate inspection method, and determining the timing of these inspections have become complex challenges due to the processes. The RBI assisted decision-makers in scheduling interventions [9].

Over the past few decades, some researchers have reviewed risk assessment approaches and their implementations in different industries. The key findings of the existing study are summarized in the following:

A reviewed article by Animah et al. [10] covers the challenges of performing a risk assessment on liquefied natural gas (LNG) facilities. These studies were classified based on risk assessment LNG applications (plants, terminals and stations, carriers and fueled ships, and offshore floating units), methods (deterministic or probabilistic; quantitative, qualitative, or semi-qualitative; traditional or dynamic), analysis tools (Bayesian network, fault tree analysis, vent tree analysis, failure mode, and effects analysis/failure mode, effects and criticality analysis), analysis output/strategy (risk-based inspection, risk-based maintenance, risk-based inspection and maintenance, facility siting, etc.), and data sources (offshore and onshore reliability data, published literature, HSE databases of UK, industry datasets, reports of regulatory agencies, and consultations from specialist). A study performed by Abbas et al. [11] presents the state-of-the-art and future directions of asset maintenance management strategies related to the repair of corroded steel structures in harsh marine environments. They investigated the application of new technologies, including artificial intelligence, computerized maintenance management systems, and Bayesian networks. Studies related to risk assessment recommendations in the form of maintenance programs were conducted by Chin et al. [12], and the recent asset maintenance approaches in the chemical industry were analyzed, including risk-based, condition-based, time-based, corrective, and opportunistic maintenance. The company employed various methods to identify the most effective maintenance approach and reliability challenges in chemical plants and integrated sites, with a primary focus on maintenance optimization.

Rachman et al. [13] carried out a review of the uses of machine learning within the context of handling and processing data in the activities of pipeline integrity management (PIM). The use of ML was investigated in several aspects of the PIM process, including inspection, observation, and maintenance, in this study. The study also examined the key elements of ML techniques, including input types, preprocessing, learning algorithms, output, and evaluation metrics applied to each PIM element. Khan et al. [5] reviewed the historical development of risk-based approaches as applied to the integrity management of pipelines. The review begins with a brief definition of integrity and its development over time. The article provides a systematic review and analysis of the available research from the public domain. The work focuses on risk-based practices in the various integrity management elements and the application of these methods throughout each phase of the pipeline life cycle.

Aditiyawarman et al. [14] reviewed the benefits of machine learning in risk management, including time-series forecasting research and an overview of risk-based inspection approaches, for example, quantitative, semi-quantitative, and qualitative. Machine learning classifiers (e.g., logistic regression, decision trees, support vector machines, K-Nearest Neighbors (KNN), and random forests) were selected to develop risk assessment tools and provide accurate predictions regarding the equipment condition and severity. This review discusses improvements to the clustering Gaussian mixture model to address the non-circular data shapes that may arise in K-means models using a deductive approach.

Despite the valuable contributions of the abovementioned studies to risk analysis, notable gaps remain. First, most articles concentrated on risk analysis with a single application or method/tool aspect. Second, few articles have comprehensively reviewed the implementation of the oil and gas sector risk analysis. Third, very few articles provided comprehensive reviews of the evolution and trends of risk analysis methods and tools.

This article systematically reviews state-of-the-art improvement optimization methods and tools for risk-based AIM in the oil and gas sector. Investigated risk-based AIM assessment methods, including quantitative, qualitative, semi-quantitative, probabilistic, deterministic, hybrid probabilistic-deterministic, and dynamic or traditional risk. Analysis tools for risk assessment and control applied in risk-based AIM, including the evolution of tools from traditional to machine learning approaches, are examined. The focus of this research is on the use of risk-based processes (risk assessment and risk control). The results of this review will enable academics and practitioners to develop a more advanced risk-based assessment model based on the challenges discussed in this article.

This systematic review is structured as follows. Section 2 outlines the study parameters, literature search methods, and summarizes the analysis of the literature. Section 3 presents a comprehensive review of selected articles covering risk-based AIM and discusses trends, improvement optimization methods, and the implementation of risk-based AIM in the oil and gas industry. Section 4 discusses future study opportunities and directions for risk-based AIM in the oil and gas industry. Section 5 concludes the paper, providing a summary of the review's main findings.

2. Review methodology

The methodological framework of this study is based on the approach of Hussain et al. and Rachman et al. [13,15]. The process comprises three fundamental steps: research definition, methodology, and analysis (Fig. 1). The following subsections describe and discuss each step.

2.1. Research definition

The target of the research definition stage is to develop the research field, define the objective, and determine the scope [13]. The research area is a risk-based approach to asset integrity management programs in the oil and gas industry. This study's objective is to examine the implementation of a risk-based approach, synthesize existing research and the accumulation of insight on this topic, and recommend future research directions. This review attempts to address the following research questions (RQ):

- RQ 1: Which journal is the most influential source in the risk-based AIM field?
- RQ 2: Who are the most active and influential researchers?
- RQ 3: What are the trends in risk-based AIM research in the oil and gas sector over the years?
- RQ 4: In which areas has the risk-based approach been improved?
- RQ 5: What risk-based AIM application is most commonly used?
- RQ 6: What risk-based AIM analysis methods are often used?
- RQ 7: What analysis tools are often used for risk assessment and control?

The main contributions of this review article are:

- A comprehensive review of existing research articles on various risk-based AIM approaches in the oil and gas sector will be conducted. The current trend in risk analysis methods and tools indicates that machine learning tools can potentially enhance traditional approaches by improving the quality of risk assessments through increased accuracy and reduced output variability.

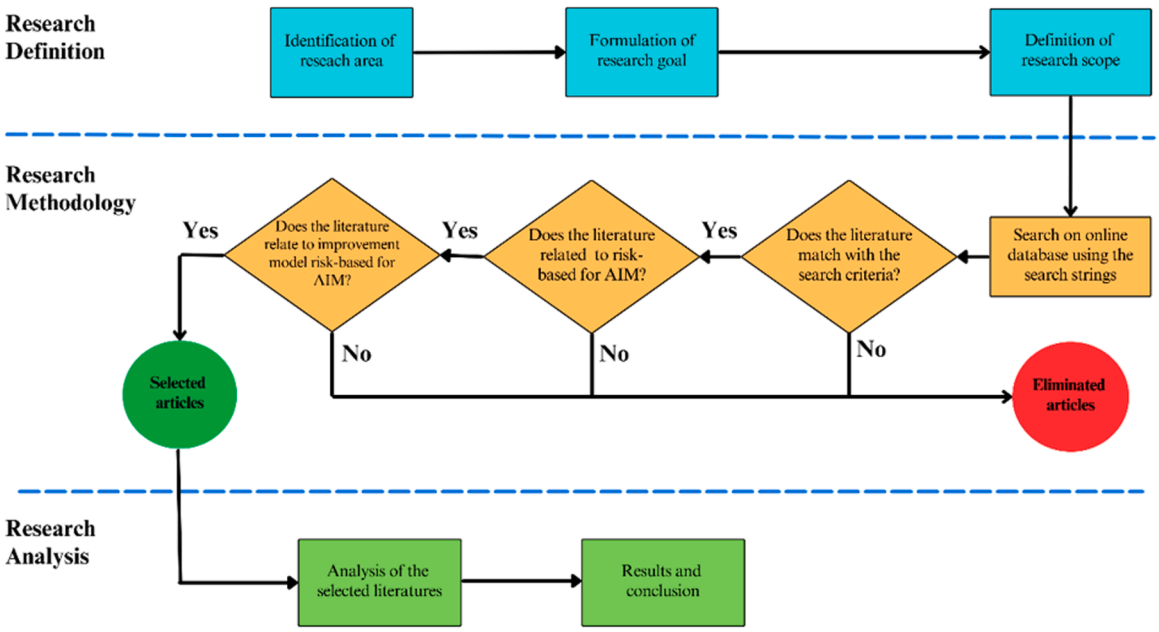


Fig. 1. Framework of methodological research adapted from [13,15].

- Analysis of existing research gaps in the current literature requires further investigation. The potential areas for enhancing risk-based AIM approaches are identified, enabling more accurate risk assessment and optimizing the risk control process.
- The future developments of the existing frameworks are also examined. Innovative elements may be incorporated or integrated into existing frameworks to enhance the effectiveness of AIM programs.

The research focus is a systematic literature review on implementing a risk-based AIM from academic research published between 2005 and 2025. This period was selected because it is a relatively recent development.

2.2. Research methodology

This work aims to conduct a comprehensive literature review in the oil and gas industry. As a result, this systematic review report was created using Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) principles and guidelines, as well as the Cochrane Handbook for Systematic Reviews of Interventions. The search strategy applied to identify relevant literature for this review is shown in Fig. 2.

The literature review was structured into four primary stages. The initial step involves gathering all relevant articles found by searching the strings in Table 1 and removing any duplicate articles. Three articles were excluded from the 1253 records found in the searched database. Following the second stage of the screening process, each article's title, abstract, and full text are scrutinized to establish whether they satisfy the specified search parameters outlined in Table 2, and the 132 remaining records continue to be assessed for eligibility. The third step was to study for the eligibility process. The result was 98 papers eligible for the review process. The fourth step is to read the entire article to achieve a holistic understanding of it and extract a dataset that will answer the research questions posed in Section 2.1.

2.2.1. Identify relevant studies databases

Studies on implementing risk-based AIM methodology are scattered in numerous publications. Therefore, the search was not focused on particular publishers. This literature review searched articles from reputable academic databases such as Scopus (scopus.com) and Elsevier (sciencedirect.com). A search for the article was conducted on August 30, 2025. Mendeley desktop (Mendeley. com), the reference manager software, managed many publications. The VOSviewer software (version 1.6.20, Leiden University, Netherlands) was used to conduct a bibliometric analysis from the author's keywords. This study does not employ a purely bibliometric analysis. The focus is on identifying the notable authors, applications, and analysis tools.

2.2.2. Search strategy and keywords

The research method uses strings and criteria to identify and select applicable academic articles. Search for related articles using a string including titles, abstracts, and keywords. The search strings were formulated using the framework of PICOC (population, intervention, comparison, outcome, and context) [16]. The population, intervention, and context components were only used because the comparison and outcome components were not relevant to the particular review. The population element includes terms related to the research topic, whereas the intervention component consists of keywords associated with the methodology or tools utilized in the study [13]. The context component includes phrases that refer to the intervention's setting [17]. The search string is shown in Table 1.

2.2.3. Inclusion and exclusion criteria

The search criteria outlined in Table 2 are specified to narrow down the retrieved articles and strengthen the reviews. The criterion, adapted from Hussain et al. [15], is as follows:

2.2.4. Quality assessment of included reviews

A systematic literature review strategy requires quality assessment to minimize the likelihood of bias associated with the searched peer-reviewed papers [18]. It is necessary to thoroughly evaluate the study's conclusions because biases induced by the study's methodology might influence the outcomes of poorly constructed research. Studies that are irrelevant to the systematic review should be obviously excluded from the systematic review or, at the very least, described accordingly. In addition, appropriate criteria for assessing the quality of the outputs and bias in the different studies need to be used.

The quality of the included studies in this review is validated using the standard quality checklist questions SQCQ adapted from Nasser et al. [19]. We included papers that answered "yes" to at least seven of the

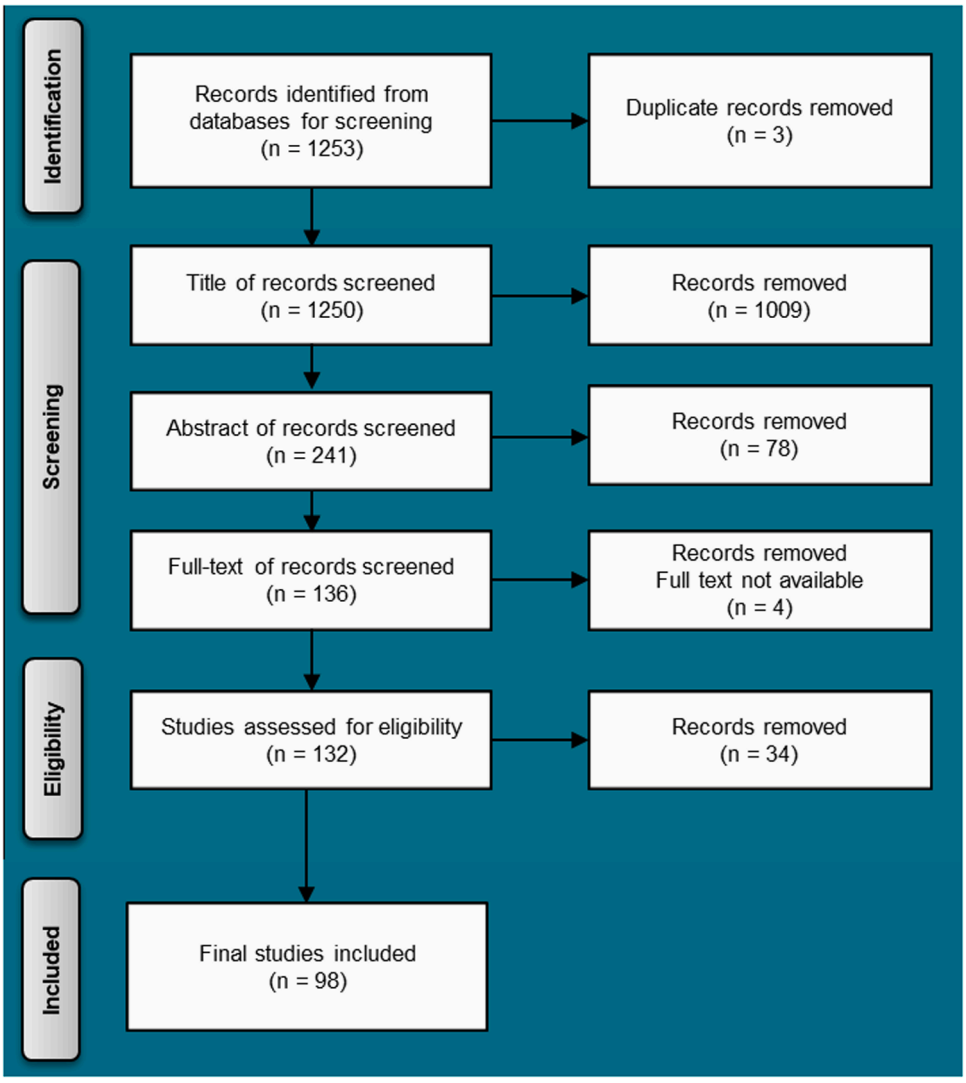


Fig. 2. PRISMA diagram indicating the systematic review process.

Table 1
Search strings to identify relevant articles.

Population		Intervention		Context
"risk-based inspection" OR "risk assessment" OR "risk analysis" OR "risk management"	AND	"machine learning" OR "deep learning" OR "algorithm"	AND	"oil" OR "gas"

Table 2
The criteria of inclusion and exclusion for the systematic review.

Criteria		Justification
Inclusion	• Articles should be published in peer-reviewed academic journals. • Studies should discuss at least the ML/DL/algorithm technique. • Studies are within the domains of risk assessment and risk management. • Articles published online from 2005 to 2025.	• Original research articles related to risk-based AIM. • Improvements compared to standard approaches • Consistent with the prescribed research scope • Recent and existing papers can help identify the fundamental and classical literature and provide a comparative analysis of a risk-based AIM period applied in the oil and gas sector.
Exclusion	• Books, proceedings, patents, graduate theses, and lecture presentations. • Articles that cannot be accessed. • Articles not written in English.	• Updated research and quality articles will be included only in the review. • To ensure that articles are available and validated. • Articles written in languages other than English were excluded.

questions. Aside from performing data extraction, the quality assessment will be analyzed in such a way that the essential findings of the studies included in the review do not have a significant influence on the review. The SQCQ employed in this study is presented in Table 3.

2.3. Research analysis

The generic risk-based AIM process is illustrated in Fig. 3. Selected articles are identified, reviewed, and categorized according to this framework.

The final selected articles were compiled to synthesize the knowledge, address research questions, and identify future research opportunities. The first research question was addressed by categorizing the journal sources of the published articles. The second research question was addressed by categorizing the researchers into first- and non-first-

Table 3
Standard quality assessment checklist questions.

No.	Standard questions
1	Is the description consistent and clear?
2	Is the purpose of the study clearly expressed?
3	Is the data collection procedure clearly explained?
4	Have the various contexts been adequately investigated?
5	Are the results valid?
6	Do the data, analysis, and conclusion connect?
7	Is the experimental process and methodology understood?
8	Is the study design well described?
9	If they are essential, are they reliable?
10	Is the study's outcome reproducible?

authors based on their contributions. The third research question was addressed by plotting the number of published articles between 2005 and 2025. The fourth research question was addressed by categorizing each article according to risk-based AIM (risk assessment and control) categories. The application was classified as a research object or case study to address the fifth research question. The risk assessment methods are outlined to address the sixth research question. The evolution of assessment tools was examined to answer the seventh research question.

As structured in Fig. 4, a framework is developed based on the components of the risk-based method for the AIM program and the improvement model. This framework is also employed to structure and summarize the articles. In this study of risk-based AIM, two main categories of the article's classification are risk assessment and risk control. Risk assessment is categorized into the probability of failure and the consequence of failure. Risk control is classified into inspection planning, process monitoring, and risk mitigation. In the analysis approach, the articles are classified into four categories: analysis method, analysis tools, application, and objective.

3. Literature review and discussion

The risk-based AIM tools referenced in the review and their associated abbreviations are listed in Table 4 to facilitate understanding of the subsequent tables. The comprehensive list of identified articles, classifications, and detailed components is based on the framework illustrated in Fig. 4.

3.1. Most influential source publications

This review only considered peer-reviewed journal articles, as specified in the search criteria outlined in Section 2.2). Studies on applying a risk-based AIM are scattered across 50 journals. The diverse scope of areas is a valid reason for the subject's interdisciplinary approach. Process Safety and Environmental Protection had the most publications, followed by the Journal of Reliability Engineering and System Safety and the Journal of Loss Prevention in the Process Industries. The top 10 most active journal publication sources are presented in Table 5, detailing the distribution of their published articles.

3.2. Most active and influential researchers

The researchers who contributed significantly and played a highly active role in the risk-based AIM were identified for observation. Table 6 depicts the most active and influential researchers of the risk-based AIM approaches for the oil and gas industry. Some productive institutions have numerous prolific researcher and their research partnerships, illustrated in Fig. 5. No identifiable grouping is in place, indicating a lack of cohesive research groups focusing on this subject. The cluster led by Faisal Khan (in green) from the Memorial University of Newfoundland and Guoming Chen (in blue) from the China University of Petroleum is currently the most active group on this topic. Other researchers outside the cluster, including T. C. Nwaoha from Liverpool Logistics, Offshore and Marine (LOOM) Research Institute and Enrico Zio from Dipartimento di Energia, Politecnico di Milano, have published three papers. Research on risk-based AIM has been conducted by various researchers and institutions, highlighting the importance of this topic, particularly in the oil and gas industry.

3.3. Trends in the application of risk-based AIMs

The relevant articles published over the years are displayed in a bar chart to illustrate the trends of risk-based AIM studies. This review focuses on academic research studies published in the last 20 years (2005 – 2025), as outlined by the research topic specified in Section 2.2. The distribution over the years illustrates the interest in risk-based AIM research methods.

The distribution of articles by publication year is depicted in Fig. 6. Nine articles were identified across ten years from 2005 to 2014 during the first period. Over the period from 2015 to 2025, the number of identified articles increased gradually to a total of 89 articles. Between 2021 and 2025, 61 articles were identified in various publications. The highest number of published articles was reached in 2024, with 15 articles. No relevant articles were recognized in 2005, 2007 - 2008, 2012, and 2014. Over the years, there has been a noticeable rise in the annual number of published articles.

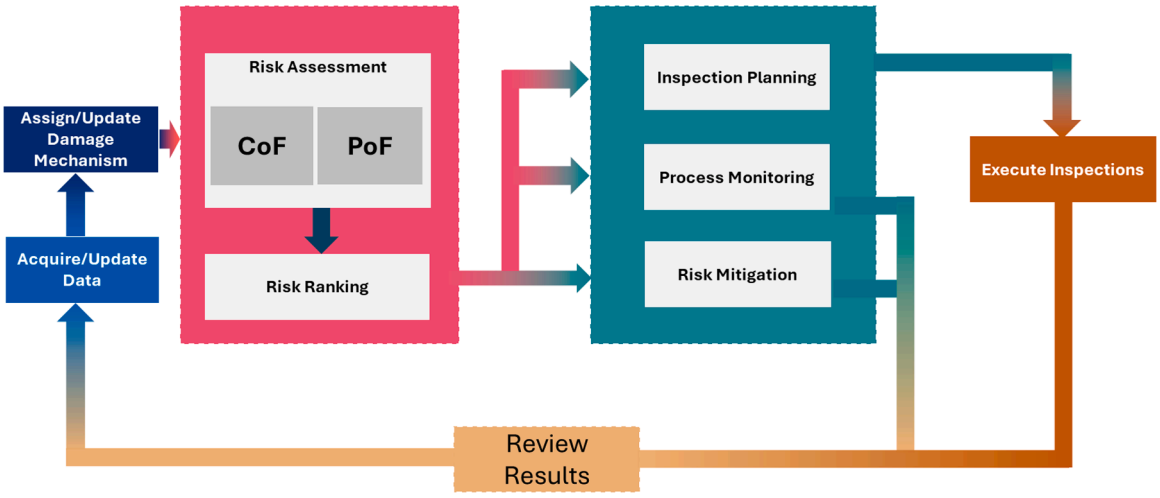


Fig. 3. Generic risk-based AIM process adapted from API RP 580 [20].

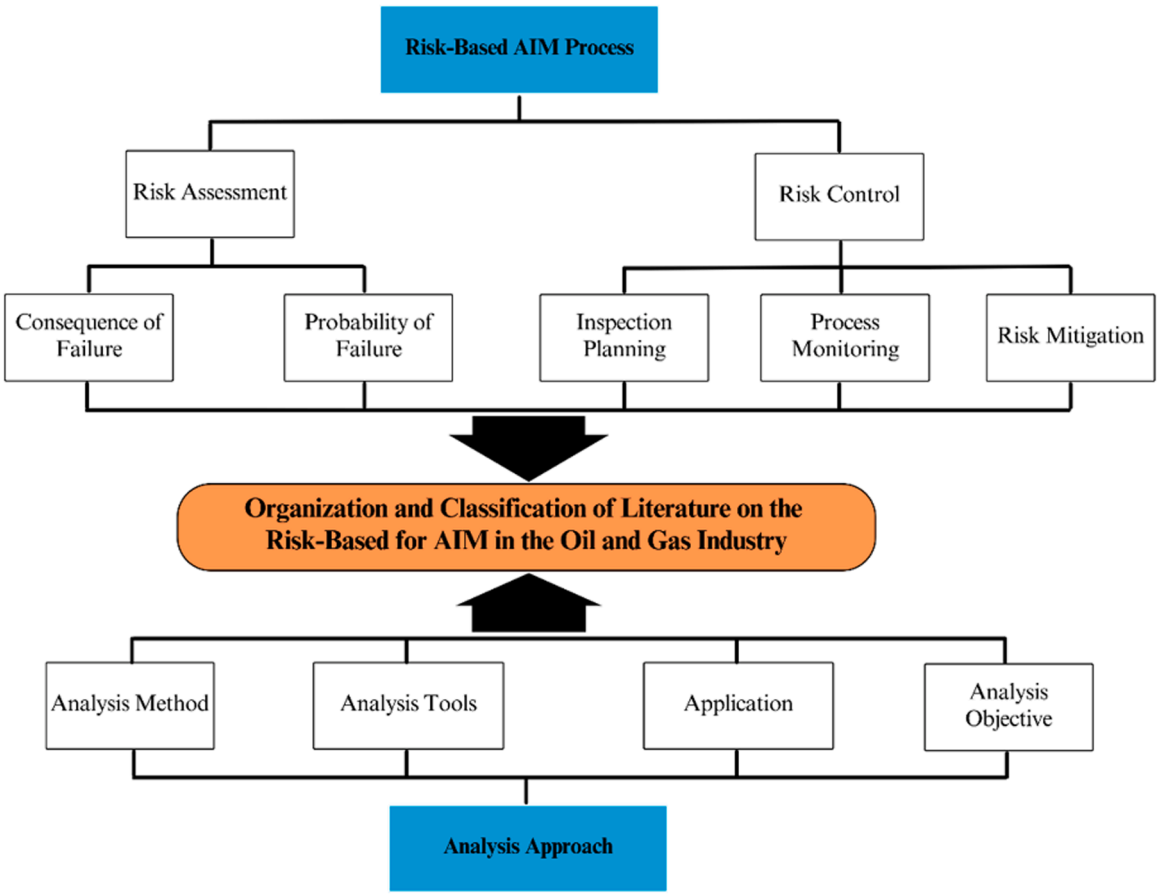


Fig. 4. Framework for the organization and classification of articles.

3.4. The research area of risk-based AIMs

The main objective of each selected study was to understand the application of improvement optimization methods in risk-based AIM. The research objective of each article is determined by a thorough reading of its entire section. The selected articles and study areas are listed in Table 7. The data in the table indicates that improvement methods were implemented in the risk assessment, CoF, PoF, inspection planning, and risk mitigation tasks. The article does not mention process monitoring. Further details about the improvement optimization for each category are given in the following subsections.

To describe each article in terms of risk-based AIM stages, including the approach used machine learning, non-machine learning, or hybrid, as illustrated in Fig. 7. This step maps the research areas that have been conducted and the research opportunities that can be developed. Previous research has focused on risk assessment, PoF, and risk mitigation. The research areas that can be explored are the CoF, process monitoring, and inspection planning.

3.4.1. Probability of failure assessment

PoF is the likelihood of an equipment or component failure due to a single damage mechanism or multiple damage mechanisms occurring under specific operating conditions [21]. The API RP 581 methodology defines the PoF as a parameter of the damage factor (DF), the generic failure factor (GFF) calculation, and the management system factor (FMS) [22,23]. By default, the GFF is a non-influential probabilistic failure that contributes to damage mechanisms developed from specific types of components according to the large population of the data component. The FMS is correlated with the adjustment of the GFF in various management process safety systems. In contrast, DF is associated with the adjustment factor of GFF, which can cause damage to the components. The author of Ref. [24] argues that the probability of failure (PoF) can be estimated using experimental burst failure pressures (BFP). For pipelines with corrosion defects, BFP is calculated using various standards (e.g., ASME B31G, DNV RP-F101) with input parameters that include pipe diameter, wall thickness, defect depth and length, yield strength, operating pressure, and ultimate tensile strength.

The presence of damage mechanisms is determined by several factors, such as corrosion, operational conditions, equipment design, and third-party damage [8,25]. Based on the review, the contributing factors that were considered in the study as input for PoF assessment are presented in Fig. 8, where the most significant factor is corrosion. Corrosion poses a time-dependent threat that significantly affects the results of risk assessment. API RP 581 outlines methods for calculating corrosion damage factors due to both external and internal thinning and the likelihood of stress corrosion cracking. Corrosion in the type of selective seam is classified as an internal or external thinning, and defects in manufacturing, construction, or fabrication may be grouped [25].

Research conducted on PoF evaluation has been divided into qualitative and quantitative methods. However, it must be understood that qualitative and quantitative methodologies are not distinct methodologies but rather two poles of a continuum. The details of these methods are discussed in more detail in Section 2.3. A qualitative approach relies on expert judgment to assess the probability of failure (PoF), drawing on operational history, planned inspections, and potential material degradation to develop relative risk scores [85]. Quantitative methods predict frequencies of failures according to data specific to the equipment. This POF can be expressed as a distribution rather than a single deterministic value and can be calculated based on the API RP 581 methodology or the failure burst pressure [22,24]. However, the qualitative and quantitative approach has some drawbacks, and it does not provide optimal results in determining the level of risk and inspection interval. The quantitative

Table 4
Risk-based AIM analysis tools and their corresponding abbreviations.

Category	Analysis tools	Abbreviation
Risk assessment	Artificial intelligence	AI
	Active learning	AL
	Artificial neural network	ANN
	API 581 Kent Muhlbauer	API-KM
	Bayesian network	BN
	Bayesian network Dijkstra algorithm	BN-DA
	Bayesian regularization artificial neural network	BRANN
	Clustering algorithm	CA
	Cascading failure chain-distributionally robust optimization	CFC-DRO
	Dynamic Bayesian network	DBN
	Dynamic Bayesian network interval Ypye-2 fuzzy sets	DBN-IT2FS
	Data fusion	DF
	Deep neural network	DNN
	Det Norske Veritas Germanischer Lloyd	DNVGL
	Digital twin machine learning	DT-ML
	Dijkstra's algorithm	DA
	Directed weighted complex network bow tie improved ant colony optimization	DWCN-BT-IACO
	Event tree analysis	ETA
	Fuzzy clustering	FC
	Failure mode, effect, and criticality analysis	FCMEA
	Finite element	FE
	Fuzzy logic	FL
	Failure mode and effects analysis vlsekriterijumska optimizacija i kompromisno resenje	FMEA-VIKOR
	Fuzzy Petri net model	FPN
	Fault tree analysis	FTA
	Fault tree analysis and human reliability analysis	FTA-HRA
	Fuzzy Bayesian network bow tie analysis	Fuzzy BN-BTA
	Fuzzy-basic probability assignment desert smarandache	Fuzzy-BPA-DSm
	fuzzy decision-making trial and evaluation laboratory Bayesian network	Fuzzy DEMATEL-BN
	fuzzy decision-making trial and evaluation laboratory fuzzy binary logistic regression	Fuzzy DEMATEL-FBLR
	Fuzzy-fault tree analysis	Fuzzy-FTA
	Fuzzy Bayesian network-fault tree analysis	Fuzzy-FTA-BN
	Fuzzy quantitative risk analysis	Fuzzy-QRA
	Fuzzy technique for order preference to similarity to ideal solution	Fuzzy-TOPSIS
	Genetic algorithm backpropagation neural network	GA-BPNN
	Gradient boosted regression tree artificial neural network	GBRT-ANN
	Graph convolutional network	GCN
	Geographic information system weight of evidence backpropagation neural network genetic algorithm	GIS-WOE-BP-GA
	Improved fruit fly optimization algorithm general regression neural network	IFOA-GRNN
	Interval-valued spherical fuzzy sets	IV-SFS
	Lévy weighted quantum particle swarm optimization artificial neural network	LWQPSO-ANN
	Multi-step backward cloud transformation sampling replacement	MBCT-SR
	Markov chain Monte Carlo	MCMC
	Modified kriging-based moving extremum framework	MKMEF
	Machine learning	ML
	Machine learning, deep learning, active learning	ML-DL-NL
	Multiphase flow-grey component analysis-frequent pattern tree-failure risk assessment	MPF-GCA-FPT-FRA
	Mean time to failure-machine learning	MTTF-ML
	Ordinal priority approach-graph sample and aggregate	OPA-GraphSAGE
	Pipeline comprehensive risk analysis	PCRA
	Persistent scatterer interferometric synthetic aperture radar	PS-InSAR
	Recursive feature elimination and particle swarm optimization AdaBoost	RFE-PSO-AB
	Recursive feature elimination cross validation-chain support vector machine regression non-dominated sorting genetic algorithm III	RFECV-CSVR-NSGAIII
	Relative risk scoring machine learning	RSS-ML
	Set pair analysis-layer of protection analysis	SPA-LOPA
	Semi-implicit method for pressure-linked equation	SIMPLE
	Support vector regression-dynamic Bayesian network	SVR-DBN
	Vlsekriterijumska optimizacija i kompromisno resenje	VIKOR
Risk control	Discrete event	DE
	Fuzzy genetic algorithm	FGA
	Fuzzy logic genetic algorithm evidential reasoning	FL-GA-ER
	Fuzzy-multi criteria decision making	Fuzzy-MCDM
	Genetic algorithm	GA
	Genetic algorithms fuzzy logic evidential reasoning	GA-FL-ER
	Genetic algorithm Pareto optimal	GA-PO
	Genetic algorithm particle swarm optimization backpropagation neural network	GA-PSO-BP-NN
	Intuitive fuzzy computational approach	IFCA
	Markov decision process deep reinforcement learning	MDP-DRL
	Machine learning	ML
	Markov model	MM
	Non-dominated sorting genetic algorithm II	NSGA II
	Ordered weight averaging analytic hierarchy process	OWA-AHP
	Risk-based inspection and maintenance	RBIM
	Risk-based inspection-multi objective genetic algorithm	RBI-MOGA

Table 5
Distribution of articles by the top 10 most active literature sources.

Journal publications	Amount	CiteScore (2024)
Process Safety and Environmental Protection	15	19.1
Reliability Engineering and System Safety	10	19.1
Journal of Loss Prevention in Process Industries	10	7.4
Ocean Engineering	6	8.4
Safety Science	4	12.8
Journal of Cleaner Production	2	20.7
Gas Science and Engineering	2	12.1
Engineering Failure Analysis	2	8.8
International Journal of Pressure Vessels and Piping	2	8.0
Scientific Reports	2	6.7

Table 6
Influential researchers and the number of studies.

Author	Number of papers	
	First author	Non-first author
Faisal Khan	–	4
T. C. Nwaoha	3	–
Enrico Zio	–	3
R.M. Chandima Ratnayake	–	3
Guoming Chen	–	3
Andika Rachman	2	–
Xiaobin Liang	2	–
Lin Fan	1	1
Huai Su	1	1
Rouzbeh Abbassi	–	2

approach incorporates more data-driven analysis, and it may be affected by missing inspection data and inspection errors due to complex operational conditions. The uncertainty and bias inherent in qualitative approaches can be mitigated using fuzzy logic, where relationships between inputs and outputs are defined by linguistic variables, offering a more realistic and versatile representation of varying conditions [88]. The semi-quantitative approach is an alternative strategy that combines qualitative and quantitative analyses, providing a good balance between precision and practicality [25]. A Bayesian framework improved the quantitative approach for probabilistic analysis and dynamic PoF assessment [29,108]. In addition, to overcome the large amount of data generated, the high dimensionality of data, usability, extensibility, and time taken for the criticality assessment technique are also relevant in its practical application. Given the limitations of traditional approaches, ML techniques represent a possible solution to the above problem by learning the mapping directly from the data [22].

3.4.2. *Consequence of failure assessment*

CoF is evaluated according to the elements that affect the severity of the hazards resulting from a hydrocarbon release, including the type of liquid within the asset and the process conditions. The consequences are classified into four consequence classifications: personnel safety and health, production losses, equipment repair costs, and environmental [8]. The CoF analysis process has been categorized as qualitative and quantitative approaches. For the qualitative method, a CoF category (such as low, medium, or high) is typically assigned for each equipment. A quantitative approach uses a logic model to characterize combinations of events that illustrate the impact of a failure. The significant factors to consider in evaluating the CoF are fires and explosions, the release of toxic substances, and the release of hazardous fluids. The proximity to the equipment where the incident occurred determines how severe the effect will be [20]. The CoF is less of a concern for performance improvement because it will remain constant [9].

Researchers proposed several quantitative methods for consequence assessment. The API RP 581 CoF assessment method, as it is in the oil and gas industry, can be easily adapted, and its application to the consequence assessment of pressurized equipment is particularly beneficial [21]. The proposed method aims to improve the traditional approach to increase the accuracy of significant factors, particularly on flammable events (fire and explosion). Qin et al. [79] predicted crater width resulting from underground explosions of natural gas pipelines, taking into account pipeline parameters, operational conditions, and installation parameters. Consequence modeling was conducted by Mousiri et al. for the actual scenario that occurred in a pressure vessel [80], and Meng et al. [99] studied the assessment of river pipeline leakage risk. In addition, the research conducted by Taheri et al. [109] that integrates real-time environment data for explosion risk assessment and safety zone applied in a pressure vessel. The ML integration allows

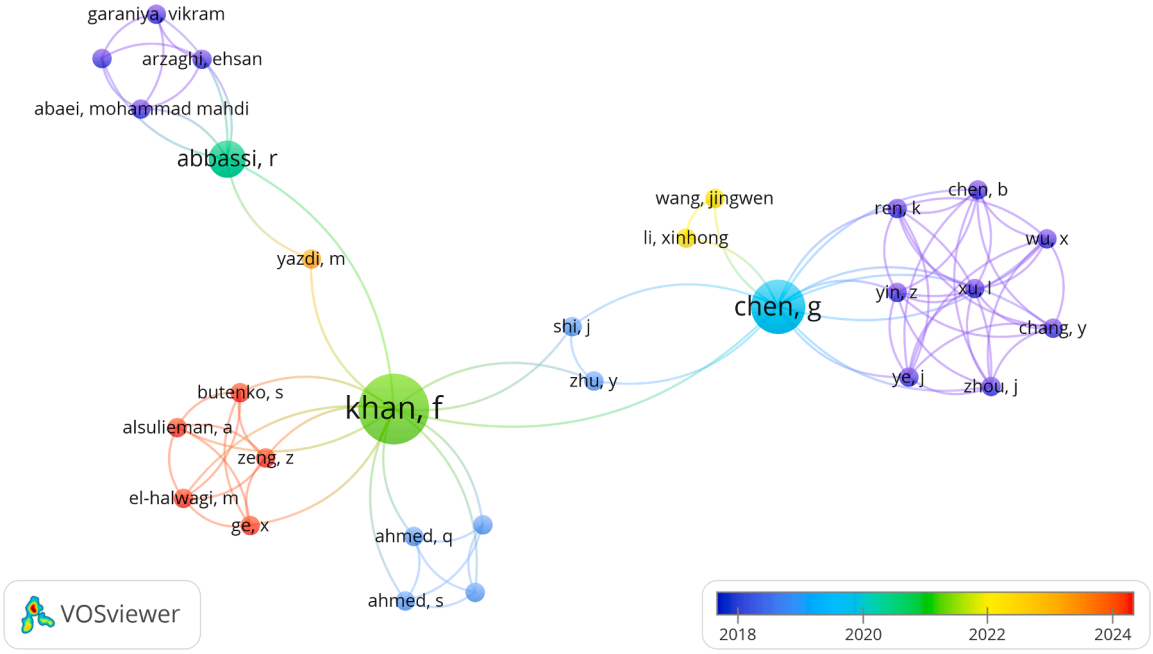


Fig. 5. Visualization of the author and their research collaborations.

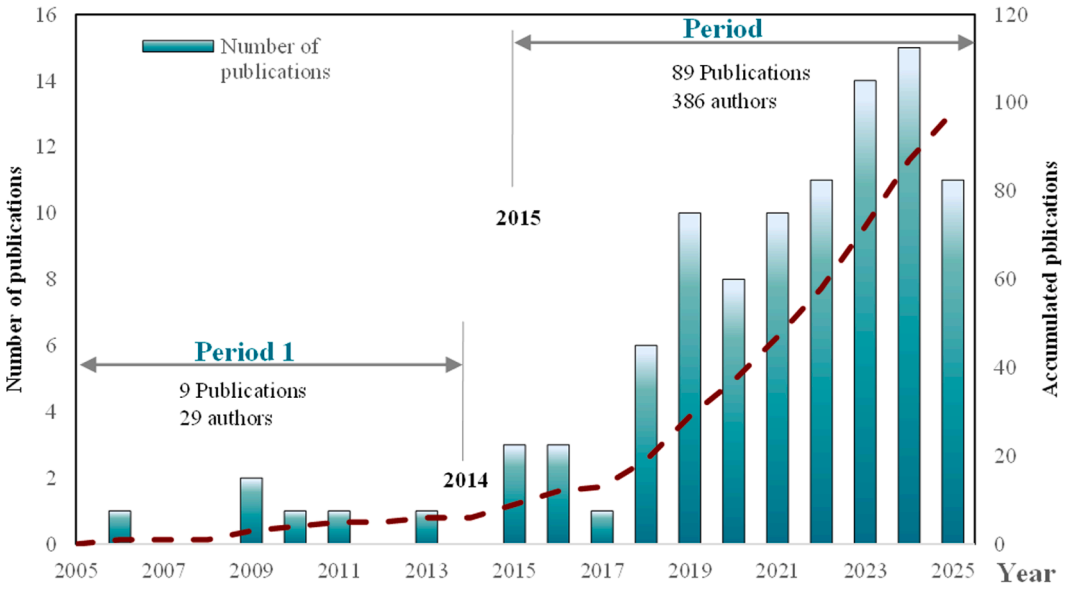


Fig. 6. Distribution of articles by year of publication.

Table 7
Distribution of the risk-based approach in the AIM domain.

Year	No.	PoF	CoF	Risk assessment	Inspection planning	Risk mitigation
2006	1					[26]
2009	2			[27]		[28]
2010	1	[29]				
2011	1					[30]
2013	1					[31]
2015	3	[32]			[9]	[33]
2016	3	[34,35]				[36]
2017	1			[37]		
2018	6	[38–40]		[41,42]		[43]
2019	10			[8,44–49]		[50–52]
2020	8	[53]		[54–59]		[1]
2021	10	[60,24]		[61–67]		[68]
2022	11	[3,69]		[25,70–76]		[77]
2023	14	[22,78]	[79,80]	[2,81–88]		[89]
2024	15	[90–94]			[23]	[95,96]
2025	11	[97,98]	[99–102]	[103–107]		

obtaining better accuracy for the CoF assessment. Zang et al. [100] predicted the possible impact radius by integrating an established numerical model of jet fires and intensity thresholds of radiation in areas of high consequence. The real-time blast load prediction in congested scenarios effectively captures the physical interactions between congestion, flame propagation, and blast load distribution [102].

3.4.3. Risk assessment

The concept of risk assessment, a key component of asset integrity management that provides a systematic process of identification, analysis, and management of risk throughout the lifecycle of an asset, has been well addressed in the literature. This practice is consistently identified in the literature as crucial for the safety of operations, regulatory compliance, and cost-effective inspection and maintenance practices. Risk-based AIM helps operators focus on the equipment that is at higher risk, which will allow for optimization of inspection intervals and allocation of resources [28,30,55]. As a result, risk assessment can be used as a decision-support tool, while also serving as an essential component to improve or build resilience into oil and gas sector operations [52,96].

Risk assessment involves a systematic use of information to identify hazards, estimate risk, provide a basis for risk assessment, and develop plans for inspection, mitigation, and acceptance of risk [20]. Indexing, matrix, and probabilistic models are the three primary forms of risk assessment, and each has advantages and disadvantages. An index model known as relative risk scoring assigns numerical values (i.e., the indices were given weights from 0 to 100; the greater the score, the better the condition) that influence risk to significant equipment conditions and activities. The causes of failure indices were the third-party damage index, corrosion index, operating error index, and design index [27,85]. The risk value of the traditional risk matrix was calculated by the probability of failure and the consequence of failure. Accuracy in risk assessment is enhanced by integrating additional models for determining both the PoF and the CoF [63,105,110]. The probabilistic models integrate the probabilities of basic events as input to estimate failure probability [60,97].

Risk assessment typically uses aspects of qualitative, semi-quantitative, and quantitative methods. For a continuum approach, these methods are considered complementary rather than competing. Qualitative risk assessment depends on engineering expert judgment and is very susceptible to bias and human error; thus, it can result in varying output assessments between different appraisers [8]. Referring to Rachman et al. [8], this variation could risk asset integrity, causing high-risk equipment to be under-inspected and low-risk equipment to be over-inspected. The weakness of qualitative risk assessment was also mentioned by Eskandarzade et al. [25], who stated that qualitative methods can not determine the most effective inspection intervals. However, several studies have improved qualitative risk assessment. Vanitha et al. [85] presented an enhanced qualitative risk assessment based on the relative risk score (RRS) using machine learning. Other enhancements were carried out by combining quantitative calculations in part of the process, especially for analyzing time-independent damage risks, often called the semi-quantitative method [25]. Eskandarzade et al. [25] proposed a procedure that combines the Kent Muhlbauer pipeline risk assessment method, which has been improved for a semi-quantitative evaluation of the time-independent damage, with the internationally accepted methodology outlined in API RP 581 [111] for a fully quantitative assessment of the time-dependent damage of the pipeline segments. Integrating these two methods offers a practical framework for developing pipeline inspection plans and controlling active damage mechanisms.

Meanwhile, developing risk assessment methods has driven the advancement of quantitative approaches. Fully quantitative RBI methods utilize mathematical, statistical, and engineering models or

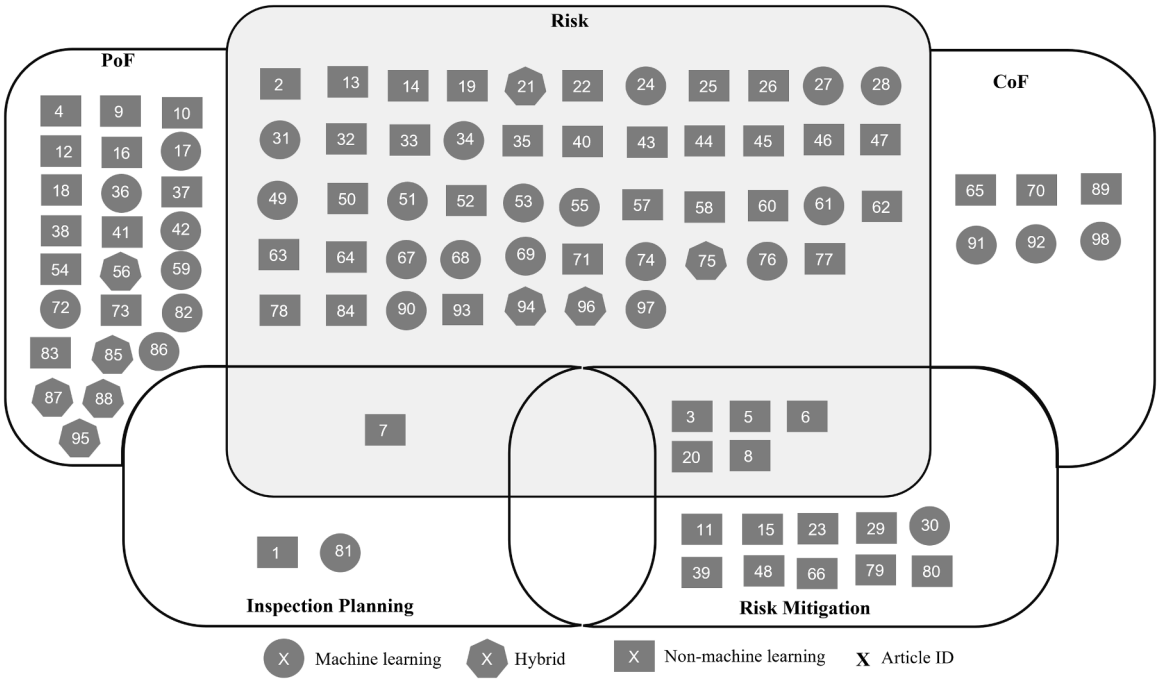


Fig. 7. Mapping of articles by risk-based steps and approaches.

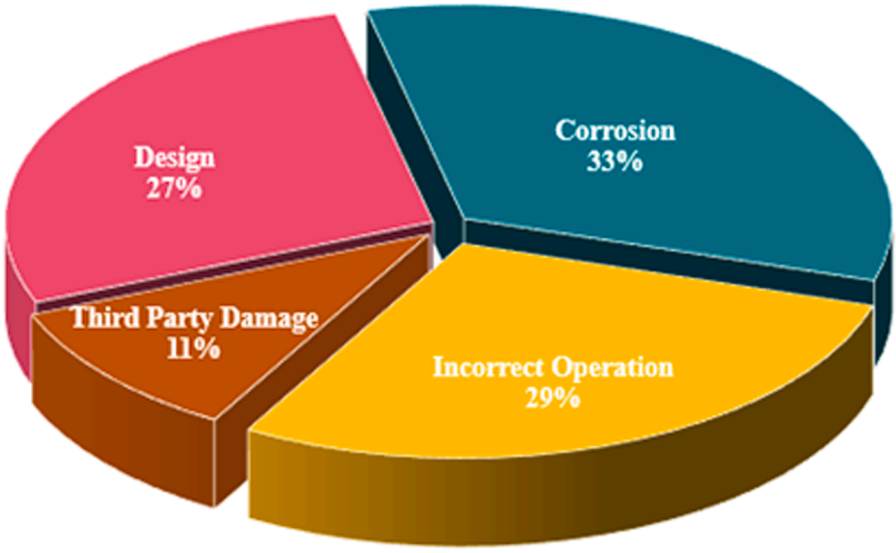


Fig. 8. Distribution of factors contributing to PoF assessment.

artificial intelligence to determine the most efficient inspection intervals [25]. A quantitative risk-based approach provides the most robust results when identifying safety-critical equipment (SCE) compared to a qualitative approach [112]. A quantitative study was carried out to examine the velocities and deformation rate of the ground during the risk assessment of underground pipelines [54]. Quantitative risk assessment (QRA) was also used by Macedo et al. [70] using machine learning in an atmospheric distillation unit.

Dynamic Risk Assessment (DRA) is defined as a process that continuously updates the risk profile of a system undergoing degradation by incorporating its current condition and the performance of protective measures. In contrast to Traditional Risk Assessment (TRA), which provides a static evaluation, DRA introduces an additional phase of monitoring and the assessment of abnormal process conditions, thereby enabling timely revision of estimated risks [66]. A DRA approach updates the calculated primary risk number considering the control system performance, safety barriers, inspection and maintenance schedules, operation protocols, and human factors [56]. Alsulieman et al. [108] studied DRA on oil and gas separator equipment, ultimately providing a more precise understanding of the system's performance. This approach enabled the early identification of potential failures and reduced risk estimation uncertainty. Dynamic Risk Assessment (DRA) can enhance Traditional Risk Assessment (TRA) by integrating its processes. Whereas, in the offline phase, TRA is utilized based on historical data and knowledge of the process; DRA, in the online phase, updates risk indices in real-time using data from condition-monitoring systems on safety-critical equipment. Wang et al. [113] proposed a model that dynamically evaluates offshore platform risks through monitoring system fault data, explicitly capturing causal relationships between risks and the severity, thereby supporting

management decisions and guiding inspection and maintenance personnel. Yazdi et al. [89] established a dynamic model to obtain the optimum solutions that enable decision makers to weigh trade-offs between reliability and the cost of integrity management actions.

3.4.4. Inspection planning

Risk-based inspection planning assumes that system risk will eventually reach or exceed user-defined thresholds related to safety, finance, or area, prompting inspection of components with the highest damage factors. Risk threshold defined by an iso-risk that the plot presents PoF and CoF on a log–log scale, with risk increasing toward the upper right, enabling clear visualization and prioritization of inspection or mitigation actions [21]. The risk acceptance criteria are typically used to decide whether planning for inspection can lower an acceptable threshold of the equipment risk of failure (RoF). Inspection reduces RoF, not by eliminating equipment deterioration processes, but by reducing the uncertainties about equipment condition, assuming the organization would respond adequately to the inspection results [8]. The effectiveness of inspection depends on the appropriateness of the selected technique and its ability to detect and characterize the specific damage mechanisms expected.

The inspection planning model must be refined to increase cost efficiency in management without underestimating risks. A proposed enhancement of the RBI program by Moura et al. [9] combines the RBI methodology with the Multi-Objective Genetic Algorithm (MOGA) that weighs costs of inspection and levels of risks while being compliant with international standards and local regulations. Unlike traditional RBI, risk is explicitly introduced as an objective to be minimized, eliminating the need for an initial risk target in the proposed RBI model. Inspection costs are also considered as an objective in that risk and inspection costs are considered simultaneously. Thus, it allows the creation of inspection programs that are optimal in terms of safety, system availability, and cost requirements. Risk or probability of failure is combined with the cost of inspection within their proposed framework to establish the ideal inspection action that minimizes risk while managing costs.

Eskandarzade et al. [25] examined the application of the refined Kent Muhlbauer method, with RBI-RP-581 offers a comprehensive framework for pipeline risk assessment by combining quantitative evaluation of time-dependent threats with semi-quantitative assessment of time-independent threats. This approach aligns closely with established industry RBI practices while ensuring complete coverage of key elements, including damage mechanism identification, failure probability and consequence analyses, and risk assessment. The optimal inspection interval date is identified by the point where the target risk line crosses the risk curve for the segment, as calculated through the quantitative RBI assessment method and considering only time-dependent damages.

Furthermore, a similar approach to the inspection planning model was presented by Campari et al. [23]. This advancement employed a machine learning technique to project fatigue crack growth in a gaseous hydrogen environment and was incorporated into traditional RBI. A method for planning risk-based inspections of hydrogen handling machinery, utilizing machine-learning techniques to predict fatigue crack growth in hydrogen gas environments. The ML technique is accurate and successfully predicts the severity of fatigue crack growth. The results of the probability of failure and consequence of failure are mapped into an iso-risk plot. Based on the defined risk threshold, estimate the inspection date before reaching the threshold.

3.4.5. Risk mitigation

Risk mitigation was implemented to the lowest level feasible by the principle of ALARP, which is “as low as reasonably practicable.” From the reviewed articles, Todinov [26] stated that risk management consists of two fundamental steps: identifying components associated with a significant contribution to the total loss of failure and directing the risk mitigation effort toward these components. Failure losses can be mitigated through three categories of measures: reducing failure occurrence, reducing failure consequences, and simultaneously reducing both occurrence and consequences.

A study by Nwaoha et al. [30] introduces a genetic algorithm-based approach to estimate the repair and maintenance costs of an LNG containment system and its transfer arm based on a total risk assessment conducted through probabilistic risk analysis. The failure rates for the basic events in the fault tree analysis of the LNG containment system and transfer arm were obtained from an extensive literature review. A thorough risk formula was created that incorporates the hazard severity weight, the frequency of system failure, and the time and financial expense involved in maintaining and repairing LNG carrier systems. This objective function is defined by a formula, with the total maintenance and repair budget as a limiting constraint influenced by the initial and unit cost of each containment system and the transfer arm’s maintenance and repair.

Hameed et al. [50] also developed a decision-making tool for scheduling maintenance based on risk in a heavily equipped gas sweetening unit within the LNG plant. The tool balances competing goals: minimizing maintenance costs and enhancing system reliability. The proposed method was tested using actual plant data, generating multiple Pareto-optimal schedules to aid decision-makers. The risk-based equipment selection strategy that is incorporated as part of the proposed decision support system will assist in reducing the production loss associated with a unit shutdown for maintenance. In addition, financial implications were evaluated.

Furthermore, Ayyildiz et al. [96] highlighted the effectiveness of risk mitigation strategies. The risk was assessed with integrated fuzzy logic and MCDM. The spherical fuzzy DM method was shown to be a suitable method for the evaluation and prioritization of such strategies. Among the most important, economic risks are the most critical, whereas social risks have been determined as the least important. The combination of Robust Design and Engineering, and Collaboration and Partnerships was the most effective at reducing the risk among those considered.

3.5. Application of risk-based AIMs

The risk-based AIM study used various case studies on oil and gas equipment. Most research areas focus on pipelines and LNG applications. The developing research areas are fixed offshore platforms, oil refineries, risers, and FPSOs. The areas with the least research are hydrogen facilities, offshore wind farms, LPG, and CCS/CCUS. The detailed application risk-based AIMs are presented in Table 8 and visualized in Fig. 9.

3.6. Risk-based AIM analysis methods

In the last 20 years, various risk assessment methods have been developed to improve risk-based AIM in the oil and gas sector. The most commonly used risk assessment approaches over the years are briefly described in Table A1. This article classifies the risk assessment approaches adapted from Animah et al. [10] based on the criteria:

Table 8
Distribution of articles by the top 10 applications.

Application	Amount
Pipeline	46
LNG	6
Simulation-based	5
Fixed offshore platform	4
Oil refinery	3
Riser	3
Hydrogen facility	3
FPSO	2
Piping	2
Pressure vessel	1

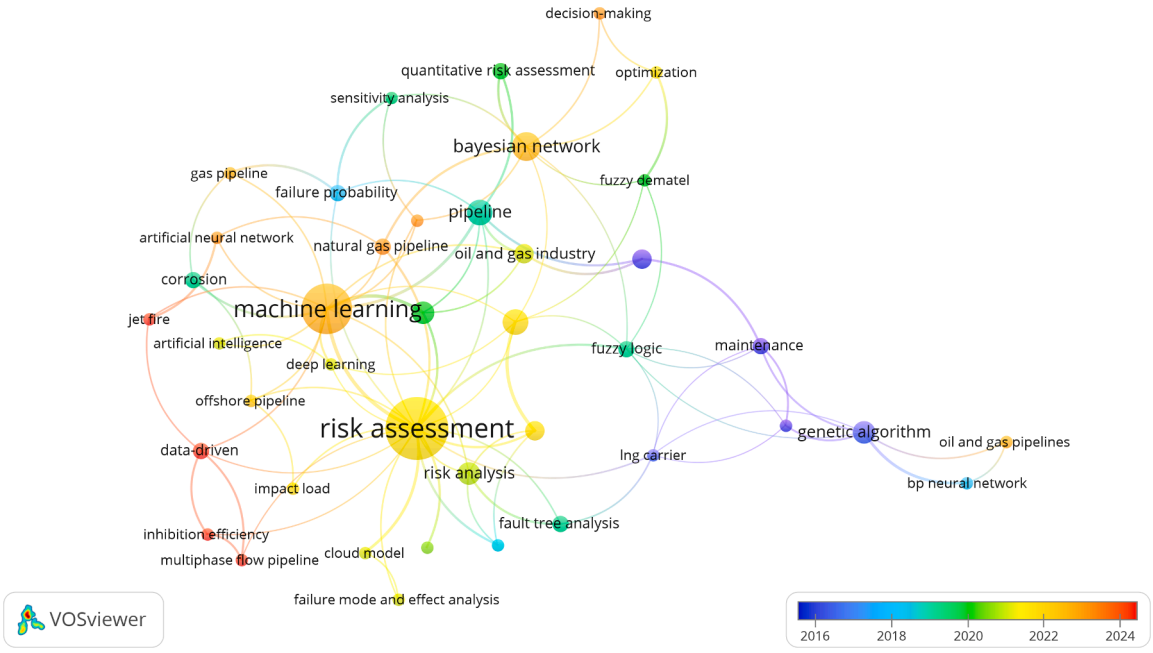


Fig. 9. Risk-based AIM research area with relevant applications and analysis tools.

quantitative, qualitative, semi-quantitative, probabilistic, deterministic, or hybrid deterministic-probabilistic; dynamic or traditional risk analysis.

The qualitative risk analysis approach is considered the most straightforward and fastest method because it depends on expert appraisal opinions and requires minimal to no quantitative or computational skills [10]. A qualitative technique relies primarily on technical expertise, rendering it susceptible to human biases, inaccuracies, and varying results among assessors. Qualitative methods do not provide optimal results in determining the level of risk and the inspection interval [25]. The example of the qualitative method could be modelled using the expert's knowledge to develop the Relative Risk Score (RRS) and categorized into risk ranking [76,85]. Due to the reliance on experts for the assessment, it is considered time-consuming and requires significant effort [8].

In contrast, Quantitative risk analysis is a structured approach employing computational and mathematical models to identify and quantify the contributing elements to PoF and CoF [56]. The quantitative approach incorporates more data-driven analysis, and it may be affected by missing inspection data and inspection errors due to complex operational conditions [47,70]. This method improves the accuracy by replacing expert judgment with the quantification of the input. According to Yan and Xu [42], the layer of protection analysis (LOPA), a semi-quantitative method, enhances risk assessment accuracy and effectiveness by replacing qualitative severity analysis with a quantitative approach. The assessment employing computational and mathematical models was conducted by He et al. [114] to evaluate the landslide risk to pipelines in various locations using a finite element model and a GA-BPNN.

To address this limitation, a semi-quantitative risk analysis integrates the principles of qualitative and quantitative risk assessment approaches. This approach overcomes the limitations inherent in each technique by integrating subjective expert judgments and quantitative risk analysis. Objective failure data are often unavailable or insufficient, limiting their usefulness in reliability and risk assessment. Research conducted by Shan et al. [38] developed a probability of failure assessment model for gas transmission pipelines incorporating historical data on failure and modification factors. This model integrates quantitative and qualitative aspects to use the available historical data and address the potential absence of precise statistical data. Eskandarzade et al. [25] proposed a semi-quantitative method for quantitatively assessing time-dependent damages from existing qualitative methods.

Risk analysis methods can be categorized as deterministic, probabilistic, and hybrid [10]. Deterministic risk analysis refers to a technique where the outputs are derived from fixed input parameters, without explicitly considering uncertainty and variability. It assumes that all input, such as defect dimension, design, and operating conditions, is known with certainty. Determining the probability of failure of pipeling using damage factor as per API RP 581 equations, where input parameters are taken as fixed input [25]. Despite being simple and computationally efficient, deterministic approaches may overstimate or underestimate risk due to their disregard for input data variability.

On the other hand, the probabilistic analysis predicts the probability of failure by considering uncertainties in input variables through risk models. The probabilistic approach is performed considering the failure events as random variables [29]. Probabilistic analysis, such as reliability evaluation and sensitivity analysis, can help to detect risks and increase the level of reliability [60]. It allows for risk quantification and confidence intervals under uncertainty by offering both a point estimate and a distribution of results. Typical methods include Monte Carlo simulation [37,53,93], Bayesian networks [29,115,116], and probabilistic fracture mechanics [24,90]. Compared to deterministic methods, probabilistic approaches offer a more realistic representation of risk but require extensive data and higher computational resources. Probabilistic risk analysis focuses on determining the likelihood of undesirable events or evaluating their probability and consequences. The PCRA method uses the relative risk model (index model) and absolute risk model (probabilistic model) to identify the most frequent failure causes of pipelines, such as release probability, wind blow direction, and probability of fatality [27]. Probabilistic risk analysis uses prior and posterior

reliability distributions to capture the time-dependent failure probability of critical components [108]. Probabilistic analyses, such as reliability and sensitivity analyses, can efficiently detect potential hazards and increase reliability levels [60]. A hybrid deterministic–probabilistic approach integrates the strengths of both methods by combining the structured rigour of deterministic models with the uncertainty-handling capability of probabilistic analysis. To overcome the limitations of deterministic or probabilistic models, hybrid models offer a solution, as deterministic models provide a structured, rigid assessment but cannot handle uncertainty.

However, probabilistic models handle variability but may lack precision when the data are scarce. Integrating both models enables more robust, realistic, and reliable risk assessments. A hybrid model was proposed by Mazumder et al. [24] to evaluate the failure risk analysis, where the input variables for pipe diameter and wall thickness use a deterministic type, while the other uses a probabilistic type. Breton et al. [29] developed a hybrid deterministic and probabilistic framework to determine the failure type of a damaged pipeline transporting a hazardous material. One of the goals of the model was to develop a reliable and straightforward process to predict a pipeline failure.

Traditional risk analysis (TRA) and dynamic risk analysis (DRA) are additional methods discussed in this review. TRA procedures have significant limitations, including a lack of high-quality data for analysis, the introduction of substantial uncertainties into the process of decision-making, the inflexibility of the approach, and the difficulty of updating overall risk assessments in response to new information, especially given the dynamic nature of complex systems [10]. Alsulieman et al. [108] described DRA as "*a method that updates the estimated risk of a deteriorating process according to the performance of the control system, safety barriers, inspection and maintenance activities, the human factor, and procedures*." Current studies have concentrated on utilizing DRA to facilitate risk-based decision-making to overcome these limitations. According to Wang et al. [86], DRA facilitates adaptation to changing environments and allows continuous monitoring of operational process risks. Assessments conducted at different time points provide valuable insights for decision-makers in effectively allocating resources. Among the benefits of bringing in condition-monitoring data is the opportunity to proactively update the risk assessment, which gives the operation team more time to apply preventive actions before an accident or a failure happens [108].

In contrast to TRA, which provides snapshots of historical data and assumptions, the DRA method adds a further stage of monitoring and assessing process abnormal conditions to update the estimated risk [66, 113]. In an oil and gas separator, DRA employs live inputs by accounting for time-dependent failure probabilities for electrical components, and the model is capable of real-time predictions [108]. Wang et al. [113] proposed a DRA model for offshore platforms using a Bayesian network model to support a dynamic Bayesian network representation of the immediate changes in key nodes associated with oil spills. The safety barrier of the safety-critical element (SCE), the pressure safety valve, can

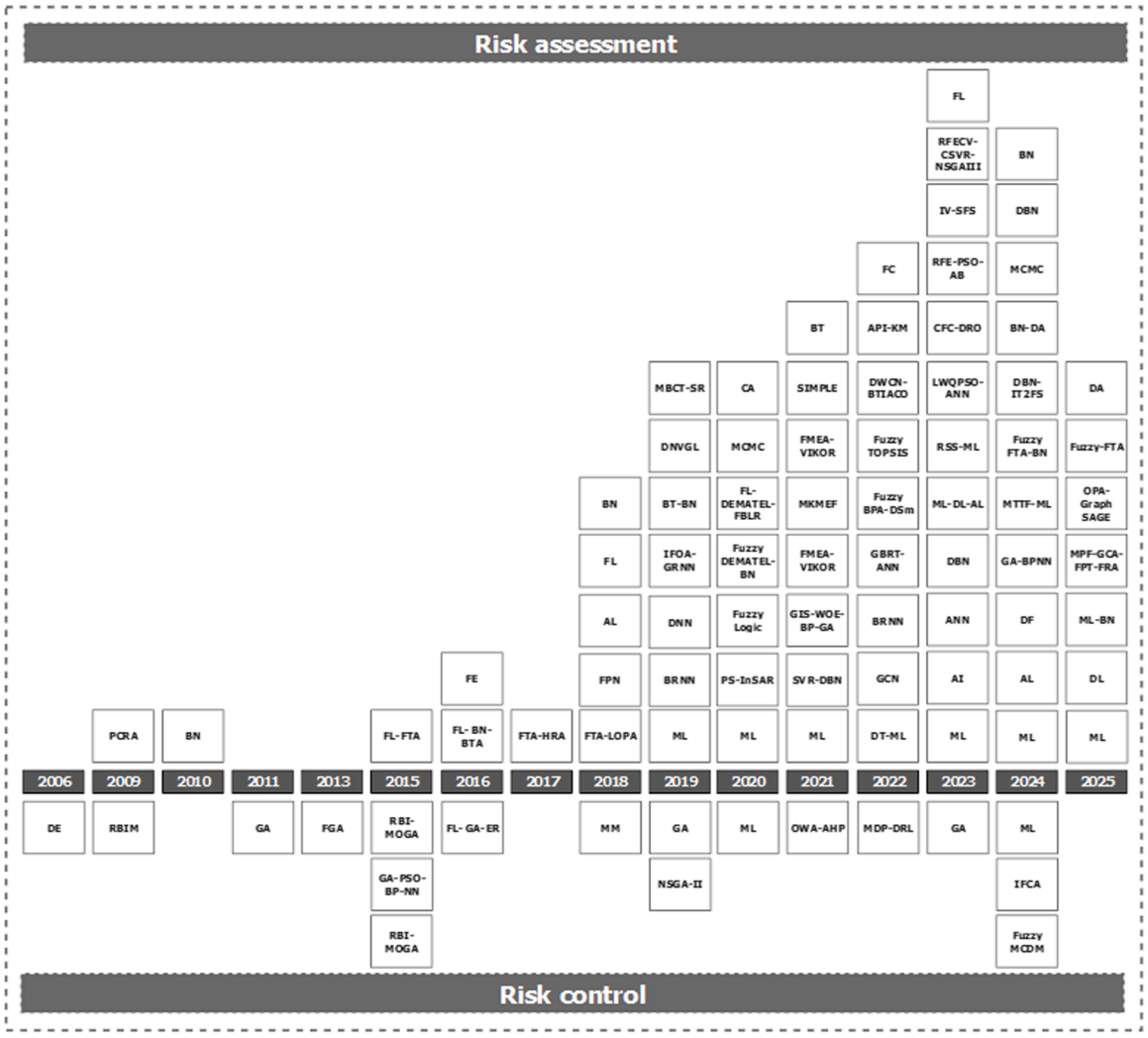


Fig. 10. Evolution of risk-based AIM tools.

be improved by DRA through integrating a data-driven logical model to predict the dynamic risk profile [66]. The notable limitation of DRA is highlighted. The reliability and limitations of the data should be tested by applying the model to independent data from the operation unit [108]. More concern should be placed upon the potential for human error in the operation phase as part of efforts toward risk reduction [66].

3.7. Risk-based AIM analysis tools

The bibliometric visualization of risk-based AIM analysis tools is shown in Fig. 9. As seen in the graph, most of the prevalent methods are part of standard methodologies, such as fuzzy sets, Bayesian networks, genetic algorithms, and machine learning. The visualization demonstrates the evolution of tools from traditional to machine learning approaches. Numerous risk analysis tools have been created in the last 20 years to aid risk-based AIM in the oil and gas sector, as shown in Fig. 10. This review describes the evolution of risk assessment and risk control tools. The following section overviews the most widely used risk analysis tools.

The findings of the risk analysis tool review are summarized in Table A1. The analysis tools showed that the majority of studies employed ML, FST, MOO, BN, RSS, MCDM, and FTA. The remaining studies used a Kriging-based moving extremum framework, SPA-LOPA, SIMPLE, Dijkstra's algorithm, MPF-GCA-FPT-FRA, distinct events, and finite element analysis tools.

To illustrate the research areas of risk-based AIM based on topics, tools, applications, and methods, Fig. 11 describes the most researched, developing, and least explored areas.

3.7.1. Machine learning (ML)

According to Simon [117], "Machine learning aims to provide increasing levels of automation in the knowledge engineering process, replacing much time-consuming human activity with automatic techniques that improve accuracy or efficiency by discovering and exploiting regularities in data." ML has the potential to streamline AIM activities through increased productivity, enabling autonomous data processing, minimizing the dependence on manual and traditional data analysis methods, and facilitating the management and evaluation of large-scale, diverse, and fast-paced data [13]. The classifications of ML approaches are depicted in Fig. 12. The descriptions of each ML domain are as follows:

- Supervised learning maps independent parameters to a known dependent parameter to infer their correlations. Supervised learning methods include logistic regression (LR), *k*-nearest neighbor (*k*-NN), decision trees, support vector machines (SVM), and neural networks. Supervised learning is divided into two categories: classification and regression.
- Unsupervised learning aims to discover patterns or hidden structures in datasets comprising input parameters and unknown output

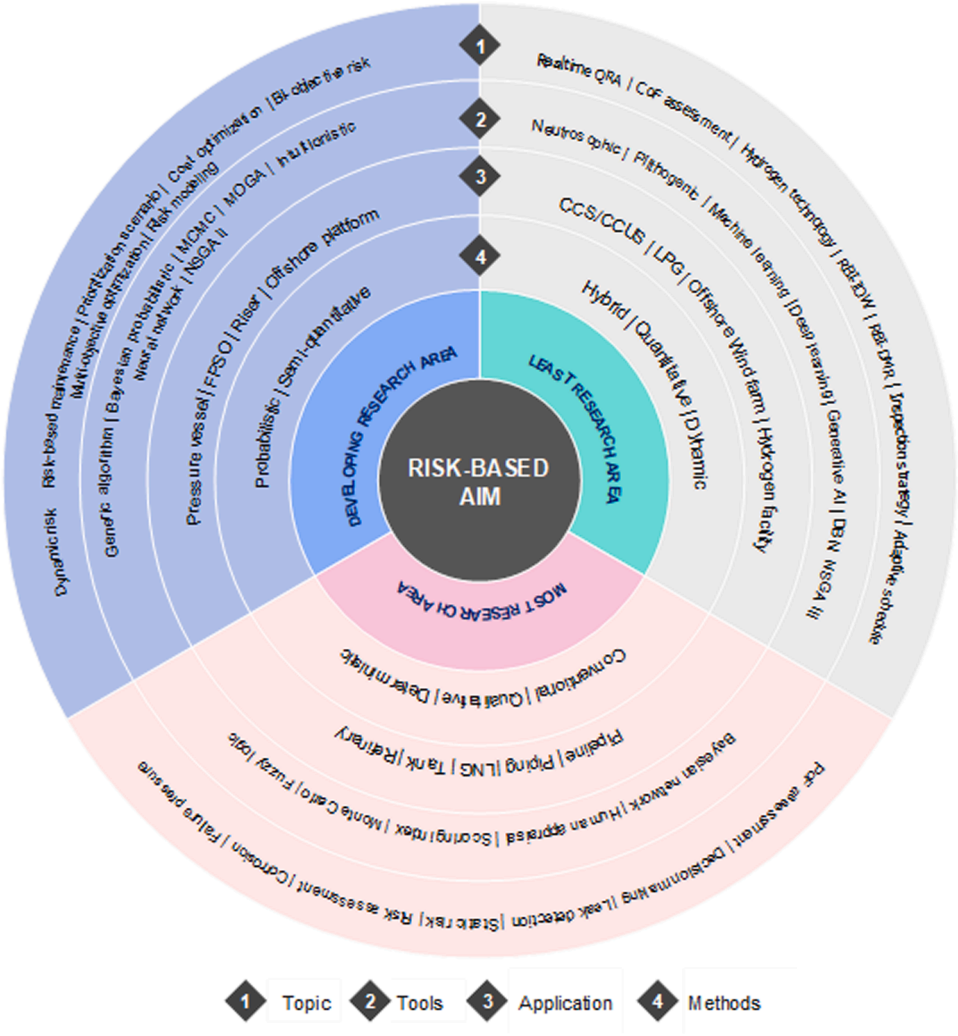


Fig. 11. The development of the risk-based AIM research area.

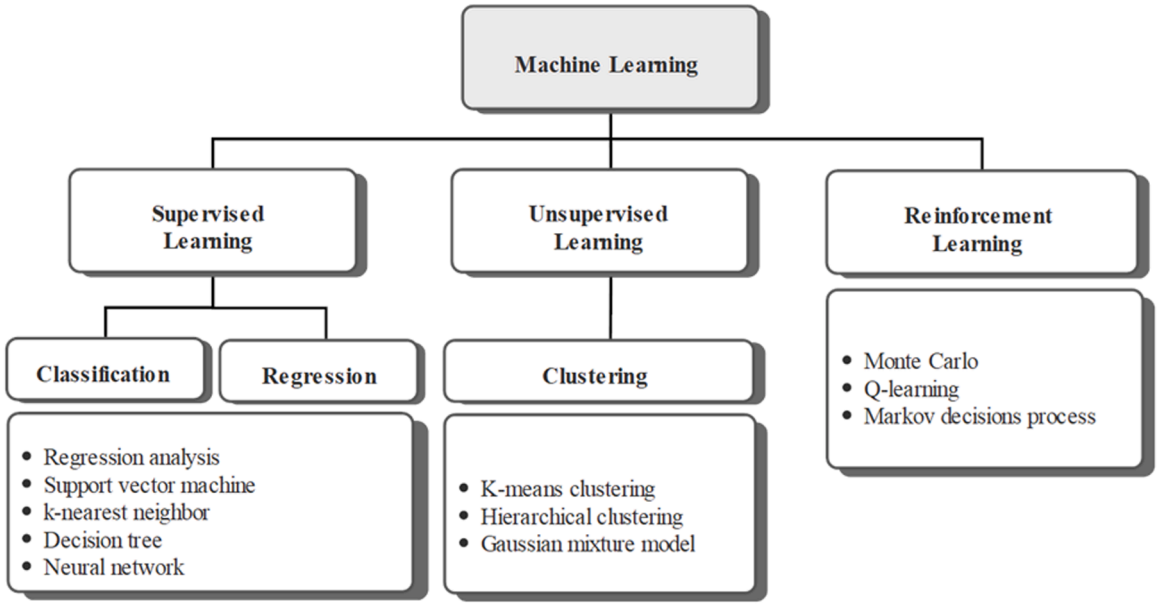


Fig. 12. The classification of ML approaches according to the form of learning feedback [13].

parameters. Clustering is an unsupervised learning process that attempts to categorize items into non-overlapping clusters. The objects in the same cluster are similar based on predefined criteria, but different from those in other clusters.

- Reinforcement learning receives feedback through rewards and punishments for behaviors. A reinforcement learning system, like unsupervised learning, does not receive datasets with known input-output parameters.

From the review, ML has been used in assessing PoF, CoF, and risk. Table 9 shows a side-by-side comparison of the ML methods used in risk-based AIM. It can be seen from the table that machine learning has been applied to various sources of data and utilizes multiple algorithms. The results of the evaluation not only highlight the models that perform better, but also the limitations of the method these algorithms employ. Machine learning has been applied to statistical data by means of classification, regression, and clustering algorithms. Multiple deep learning architectures have been proposed to handle more complicated datasets as well. Support Vector Machines, *k*-Nearest Neighbors, and Random Forest are the top three algorithms in popularity amongst the machine learning algorithms. The Random Forest classifier is especially popular because it has high prediction accuracy and is computationally efficient [78].

For probability of failure (PoF) analysis, machine learning approaches leverage larger and more diverse pipeline datasets as data sources. Machine learning algorithms have been utilized to analyze the probability of failure associated with damage mechanisms, particularly those induced by corrosion. The study indicated that machine learning algorithms perform promisingly and can serve as an effective alternative to classic physically based approaches, whether deterministic or probabilistic, for predicting failures in oil and gas pipelines [24]. The ML model predicts corrosion areas with more precision, demonstrating its validity and suitability for consistently reflecting the actual results of in-line inspection [22]. The machine learning models developed were efficient and accurate in predicting the failure pressures of corroded pipelines [90]. However, some drawbacks were discovered, most notably the use of a single corrosion defect without accounting for the presence of many defects and their potential interactions, which could diminish residual strength and increase the associated failure risk [93]. ML shows that modeling integration with human appraisal remains beneficial in predicting PoF [22].

Consequence of failure (CoF) analysis incorporates a wide range of datasets, including those from offshore platforms, pressure vessels, and pipelines. Machine learning algorithms have been utilized to analyze the Explosion Risk Analysis (ERA). A TBRANN model has been proposed for dispersion studies on offshore platforms, effectively capturing the transient behavior of flammable cloud volumes [49]. A deep spatio-temporal sequence model with an attention mechanism is proposed to capture methane diffusion dynamics, enhancing the extraction of key leakage features while reducing computational complexity [103]. The ML model can quickly predict the flame length and radiation intensity of Hydrogen-blended natural gas (HBNG) pipelines with jet fire [101]. Critical recommendations were emphasized. Sole reliance on computational methods is not a rational choice in risk assessment; expert judgment remains essential for interpreting risk scenarios and complementing data-driven analyses [75]. The Physics-GNN approach lacks explicit uncertainty quantification, which limits its applicability in decision-making contexts requiring confidence estimates or in scenarios with inherent uncertainties [102].

Risk assessment is carried out for equipment in upstream and downstream facilities. The application of machine learning in risk assessment incorporates both the PoF and CoF. Risk screening assessments are largely qualitative and heavily reliant on engineering judgment, making them susceptible to human bias and error. The application of machine learning mitigates this issue by reducing subjective influence and minimizing appraiser-to-appraiser variability [8]. Probabilistic risk assessment is another contribution of machine learning applications, addressing the complicated and nonlinear interactions between simple events and their outcomes. Compared to traditional procedures, this approach is faster and more efficient because it is data-driven [71]. Validation using data from offshore platforms and a production separator leak incident demonstrates that the suggested text-mining

Table 9
Comparison of the ML methods used in risk-based AIM.

Category	Ref.	Data source	Algorithm	Outcome/ findings
Probability of failure	[53]	Offshore pipeline	MLP, RBFN, SVM, GP	The GP model showed the best performance. Failure probability decreased significantly with greater burial depth.
	[60]	Pipelines	KNN, DT, RF, NB, AB, LB, CB	The study considers external corrosion only. The XGBoost is the optimal machine learning model.
	[22]	Pipelines	LR, kNN, SVM, ANN, RF, GB, AB	SML extracted key risk information from USML. SVM and LR outperformed other ML models. DL with human appraisal remains beneficial for risk assessment.
	[78]	Oil and gas pipelines	GAN, RF	GAN generates synthetic data. RF validates failure risk. RF is effective with low computation. Burst failures in pipelines stem from active corrosion.
	[90]	Gas pipelines	ANN, SVR, KNN, DT, RF, AB, GBDT	ML and empirical models compared. ANN shows superior performance. DNV RP-F101 outperforms other empirical models.
	[93]	Gas pipelines	ANN, SVM, KNN, DT, XGB, GBM	Analysis considers a single corrosion defect. ANN is the top performer. The use of SHapley Additive exPlanations (SHAP) improves model interpretability.
Consequence of failure	[49]	Offshore platform	TBRANN	The model provides more accuracy than the Frozen Cloud Approach (FCA) for the transient dispersion study.
	[75]	Urban Pipeline Network	GCN	Good scalability and efficiency. The model lacks pipeline-level evaluation and clear clustering standards. Expert judgment remains crucial for risk decisions.
	[100]	Gas pipelines	BPNN, CNN, RF	The potential impact radius increases with increasing pipeline pressure and diameter. BPNN provides the best prediction performance.
	[101]	Gas pipelines	BPNN, CNN, RF	The proposed model improves prediction and extends applicability from <2 to ≤ 12 MPa. CNN shows strong predictive performance.
	[102]	Semi-submersible platform	Physics_GNN	Considering the interaction between the risk factors. Offers computational speed. Limited knowledge transferability. Lack of uncertainty estimation.
Risk assessment	[8]	Pressure vessel and piping	LR, SVM, k-NN, GBDT, AB, RF	Comparative study between a human appraiser and a machine classifier. Ensemble techniques (GBDT, RF, and AB) perform better than single classifiers (LR, SVM, and kNN)
	[57]	Urban Pipeline Network	SC, HC, DBSCAN	Fully data-driven unsupervised risk assessment. The clustering model enables standard risk level assignment.
	[70]	Atmospheric Distillation Unit	SVM, MLP, AB, RF, KNN, GBDT	Classification of potential accident scenarios. ML is applied in the initial stage of QRA. GB predicts likelihood, SVM predicts severity.
	[71]	Subsea gas pipeline	BRANN	Dynamic risk assessment captures nonlinear, non-sequential accident

(*continued on next page*)

Table 9 (*continued*)

Category	Ref.	Data source	Algorithm	Outcome/ findings
				escalation. Risk factor probabilities are assumed due to data scarcity. Model updates with new data to reassess accident probabilities.
	[74]	Oil pipeline	SVM	Prognostics estimates remaining useful life (RUL) by detecting failure precursors from pressure-based risk conditions.
	[82]	liquefied petroleum gas (LPG) terminal	ANN	Threat zone distances predicted for LPG leaks. Consequences are estimated only in extreme summer and winter. There was no mention of risk mitigation measures.
	[2]	Offshore platform	BERT-BiLSTM-CRF	The proposed model provides accurate risk identification factors. Risk assessment simplifies the use of hidden danger data.
	[118]	Offshore pipelines	SVR, KNN, RF, GBR	The GBR model shows insensitivity to data distribution. The GBR model exhibits greater robustness and stability.
	[23]	Gas pipelines	GBM	Data imbalance and crack growth reduce Precision and Recall for high severities. ML predicts Fatigue Acceleration Factor and severity class.
	[103]	liquefied natural gas (LNG)	1D-CNN–BiLSTM–AM	Interpretive structural modeling identifies leak response and auxiliary measures as key decision drivers. The model predicts leak risk with higher accuracy.
	[107]	Natural gas station	IGM-ISVM-IPOA	IPOA-optimized combined model achieves higher prediction accuracy than separately improved models.

approach efficiently streamlines the risk assessment procedure, effectively uses field data, and correctly detects risk variables [2]. Notable suggestions were brought forward. The accuracy of ML will largely depend on the adequacy of the data. If the practical data is available, the model can be further enhanced [71]. In addition to risk assessment, ML was used in planning inspections to predict mechanical fatigue due to hydrogen based on material properties and operating conditions, allowing vulnerable components to be more accurately identified. Such predictions help adapt inspection periods and policies to intervene on time in hydrogen transport systems [23]. Although the existing literature on risk assessment leans towards a probabilistic focus, it should incorporate a quantitative evaluation of consequences to form a more complete and balanced framework for the planning of inspections. Future research should focus on approaches that integrate hidden danger data for consequence evaluation [2].

3.7.2. *Fuzzy set theory (FST)*

Fuzzy set theory is one of the well-known engineering approaches to risk that deals with the uncertainty, vagueness, and subjectivity of expert opinion [32,88]. While classical probability theory is based on crisp data, fuzzy logic allows the use of imprecise linguistic variables such as “low risk, moderate risk, high risk”. This characteristic makes it especially appropriate for cases where historical failure data are incomplete. Fuzzy-set theory was introduced as an effective way to address the difficulties inherent in traditional risk assessment techniques [10]. As clearly shown in Fig. 13, classical and fuzzy sets for the ‘safe state.’ While classical sets impose a rigid boundary between safe and unsafe, fuzzy sets allow a gradual transition, reflecting that safety is inherently uncertain and cannot be strictly dichotomized. Fuzzy logic, through membership functions, quantifies this uncertainty and has been effectively integrated into process risk analysis to reduce knowledge gaps.

Fuzzy set theory has been used as part of a risk-based asset integrity management. This process of translating expert knowledge into fuzzy membership functions allows for the systematic quantification of risk factors even when datasets are incomplete. Furthermore, the Fuzzy-set theory is often linked to traditional risk assessment methods, resulting in fuzzy-based approaches to assess the risk and dependability of intricate systems in high-risk sectors. Fuzzy hybridizations, such as fuzzy fault tree analysis [32], fuzzy Bayesian networks [35,116], fuzzy clustering [76,81], and fuzzy genetic algorithms [31], have proved successful in producing better predictions and improving decision support, indicating that fuzzy set theory can serve as a solid starting point in the development of risk assessment.

3.7.3. *Multi-objective optimization (MOO)*

MOO is a promising approach that has been developed to deal with trade-offs between conflicting criteria in engineering design decision-making. In some cases in risk-based asset integrity management, decisions are made by balancing safety against cost, reliability, and inspection intervals. Traditional single-objective methods are insufficient to reflect these competing priorities and can result in suboptimal or biased outcomes. Research conducted by Yazdi et al. [89] reported that two algorithms can solve the optimization model. The first approach involves exact optimization methods, such as the Branch and Bound algorithm, which provide precise solutions. The second approach uses meta-heuristic algorithms, including genetic algorithms, grey wolf optimization, bee colony optimization, whale optimization, and ant lion optimization. Optimal multi-objective functions are typically achieved

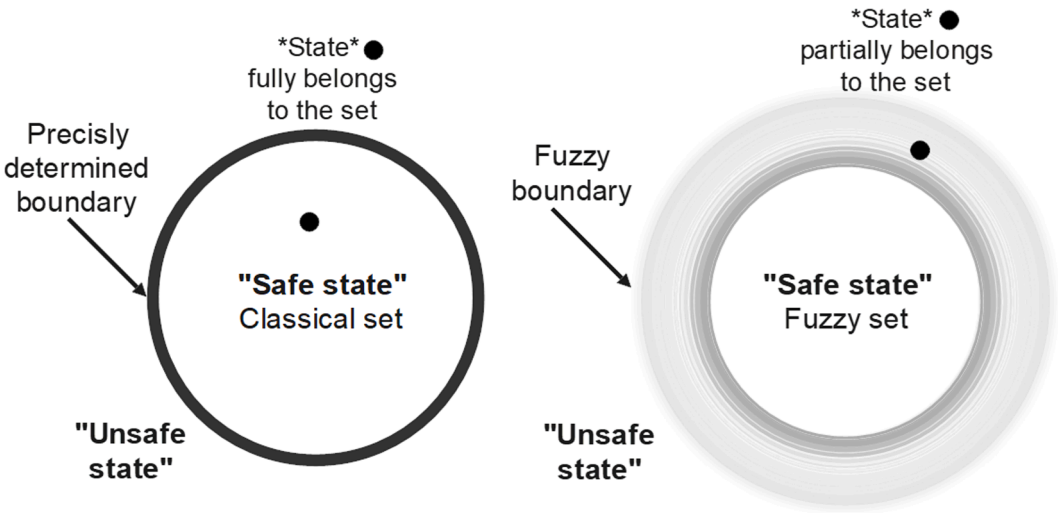


Fig. 13. Classical set and fuzzy set for safe and unsafe states [58].

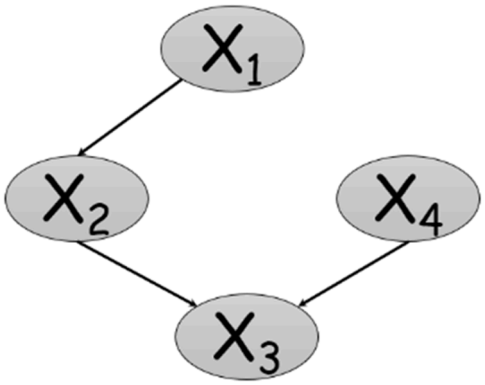


Fig. 14. Simplify the schematic of a Bayesian network.

using enumeration or random approaches that align with the capabilities of meta-heuristic algorithms. High-level meta-heuristic algorithms are employed to develop, locate, or select heuristics (particular search algorithms) that successfully tackle complex multi-objective optimization problems. The proposed framework can be based on decision-making support tools for optimal maintenance of a corroded subsea pipeline subjected to risk, safety, and resource integrity management.

A stochastic method, genetic algorithm (GA), is chosen for finding optimal solutions because of its robust effectiveness and ability to conduct efficient global searches [9]. The analysis tool of a multi-objective genetic algorithm (MOGA) aims to mitigate the probability of failure and the costs associated with it while adhering to constraints defined by local regulatory requirements and/or international standards. Furthermore, Moura et al. [9] described the RBI+MOGA approach as offering several advantages: (i) no need for a user-defined risk threshold; (ii) estimation of failure consequences is not required; (iii) inspection costs become more manageable, thereby facilitating the evaluation of the effect of investments in protection on the levels of risk; and (iv) a model inherently suggests how to allocate the inspection budget efficiently as part of its solution. An article reported by Su et al. [51] presents the nondominated sorting genetic algorithm (NSGA) II approach, which has been selected to achieve good results when optimizing natural gas pipeline network operation, particularly in ensuring reliable supply and enhancing operational efficiency.

3.7.4. *Bayesian networks (BNs)*

Bayesian Networks are now a popular probabilistic modeling technique for capturing complex engineering systems that exhibit dependencies and uncertainty. More than Traditional fault trees or event trees analyses, BNs also represent causal links between risk factors and dependencies conditioned on other risk factors, allowing for a more dynamic and transparent reasoning under uncertainty. Bayesian networks are based on Bayes' theorem, which is presented as a graphical, mathematical representation of the correlation between nodes and variables. Each variable consists of a collection of states that can be true or false. Every node is linked to a unique probability distribution. An article reported by Chen et al. [40] provides a simplified schematic of the BN, as shown in Fig. 14, which presents a BN in which node X_2 is represented as the child node of X_1. X_1 and X_4 are considered root nodes since they have no parent nodes, and nodes with no children are termed leaf nodes. In contrast to the classic bow-tie technique, which arranges interactions between random variables in a chronological sequence, BNs model the interdependence of variables, whether continuous or discrete, over time.

In risk-based AIM, BNs have been employed to estimate the probability of failure, update risk assessments with new inspection or monitoring data, and model degradation processes. Their ability to incorporate both quantitative data and qualitative expert judgment makes them particularly valuable when failure data are scarce or incomplete. The BN model was used for a dynamic risk assessment to evaluate the risk of offshore platforms through the input evidence (fault data) of the monitoring system and intuitively reflect the causal relationship between risks and the severity of risk occurrence [113]. A Bayesian probability-based method was used to determine and predict the failure type (rupture or leakage) in steel pipes operating in actual corrosion conditions [29]. Recent research extends BNs applications through hybrid models, such as fuzzy Bayesian networks and dynamic Bayesian networks [86,115], Bayesian networks and machine learning [49,77,94], Bayesian network and Dijkstra algorithm [119], Bayesian network and fuzzy fault tree [116], which enhance flexibility in handling linguistic uncertainty and time-dependent risk evolution. These advances position BNs as a core methodology for intelligent and adaptive risk analysis frameworks. BN effectively addresses the issue of dynamic risk assessment by accommodating changes in input

parameters and recalculating associated probabilities [119].

3.7.5. Relative risk score (RSS)

RSS is an index-based methodology that uses numerical scores to assess primary equipment conditions and activities, thus contributing to the total risk assessment. The scoring algorithm comprises multiple hierarchical layers based on specific assessment requirements. Factors related to failure or consequences are analyzed, and their relative contributions to risk, consequences, or overall risk are evaluated to determine the most effective strategy for minimizing risk [85].

Several studies have sought to enhance the Relative Risk Scoring (RSS) model to improve its reliability in pipeline risk assessment. For time-independent damages, researchers have refined Kent Muhlbauer's methodology by introducing logical weighting factors and modification parameters to represent diverse operating conditions better. In contrast, time-dependent damages such as corrosion are often evaluated using the general failure frequency method recommended in API RP 581, providing a probabilistic foundation for risk estimation [25]. Recent developments have incorporated machine learning approaches within the RSS paradigm, allowing for adjusting weights of the risk factors, finding patterns in the inspection data, and building predictive models of failure [85]. In some of these hybrid methods, the subjectivity of traditional RSS is minimized, and data-driven accuracy, as well as risk-based decision-making, can be utilized.

3.7.6. Markov chain model (MCM)

Markov chains, which constitute a powerful tool for reliability engineering, are used to model systems that transition between discrete states at each time step with specific probabilities. Tools based on MCM are suitable for assessing the risks associated with dynamic systems, as they include advanced quantitative methods that employ diagrammatic models to evaluate the movement between different states. The system's state evolves according to a specified probability distribution. This tool has a dependency structure, wherein the probability of future states depends solely on the current state, without any influence from past states [10].

The MCM method has been used for several studies related to risk-based AIM. According to Redutskiy [43], a Markov model represents device failures, repair processes, and technological incidents to describe the performance and maintenance of safety instrumented systems. Recent research extends MCM applications through a hybrid with other models. A stochastic capacity network model is coupled with a Markov model and graph theory to generate the state space of the pipeline network system. An integrated model of maximum flow and a Markov model is developed to simulate the maintenance scenarios for the DRL-driven maintenance strategy [77]. Dynamic risk assessment was studied in oil and gas separators, with critical safety barriers selected for condition monitoring. During the online phase, the condition-monitoring information is applied to perform real-time updates of the safety barriers' reliability by means of a sequential Markov Chain Monte Carlo (MCMC) algorithm driven by a BN. The updated reliabilities of the safety barriers are, then, used in the offline risk assessment model for a DRA [108]. A hidden Markov model (HMM) is used as an extended classical Markov model for a risk estimation model [92].

3.7.7. Multi-criteria decision-making (MCDM)

The use of MCDM methods is prevalent in the engineering field of risk management, which involves conflicting objectives and sets of noncommensurable criteria. The criteria can be quantitative or qualitative, which poses challenges to the Traditional single-metric approach.

Table 10
Summary of future research opportunities to enhance the risk-based AIM method.

Topic	Ref.
Dynamize the model	[102,106,51]
Incorporate continuous variables and data mining methods	[2,47,97]
Expand data availability and quality	[70,79,87,90,101,103,105,120]
Validate and optimize model structures	[8,89,93,94,119]
Test under diverse scenarios	[38,55,62,107,118]
Collaborate with industry and validate through real systems	[24,96,98,71]

MCDM combines technical, economic, environmental, and safety considerations into an organized framework that allows for systematic and transparent decision-making in complex situations [10].

Some articles related to the MCDM and their advancement with other tools were identified in the literature review. A combination of OWA-AHP methods for risk analysis of land suitability preparation, where AHP weighs each parameter, and OWA considers risk levels. It means the AHP method is effective enough in such decision-making problems, and, at the same time, the OWA method prepares a decision-making framework according to different risk levels [68]. Various MCDM techniques have been applied to rank the priority of risk associated with detected failure modes to address the limitations of risk priority number (RPN)-based FMEA [62]. The TOPSIS method ranks all solutions within a given set using negative and positive solutions for a multi-objective problem [72]. A modified FMEA approach based on a cloud model and extended VIKOR is proposed to enhance the effectiveness of the FMEA in risk assessment under uncertainty [62,64].

3.7.8. Fault tree analysis (FTA)

FTA is a widely used deductive framework for systematically identifying and analyzing the combinations of component failures and external events that can lead to system-level hazards. By representing the logical relationships between basic events and top events through AND/OR gates, FTA enables the quantification of failure probabilities and the identification of critical contributors to system risk. A key benefit of FTA is its capacity to assist decision-makers in identifying a system with the most essential components, thereby allowing them to develop the most effective maintenance strategies [10]. A Traditional static fault tree approach has a limitation in that it cannot explicitly account for uncertainties in failure data, leading to decreased accuracy [116]. The general FTA process is: system understanding, accident investigation, top event determination, event tree construction, event tree optimization, qualitative or quantitative analysis, and development of safety measures [37].

The FTA and its advancement with other tools have been used for several studies related to risk-based AIM. The integration of FTA and BN improves traditional BN by enabling fuzzy logic relationships between nodes and facilitating conditional probability table (CPT) construction from limited or uncertain fault probability data. It also overcomes the limitations of traditional FTA, such as cumbersome operations and a lack of bidirectional reasoning, thus advancing fault tree and BN analysis [116]. The FTA–based method, enhanced with AHP and fuzzy theory, overcomes traditional FTA limits by enabling probabilistic analysis of pipeline leak–induced domino effects for improved pipeline risk management [104]. Integrating FTA with HRA via Monte Carlo Simulation improves risk assessment reliability and guides LNG carrier management [37]. Traditional FTA was enhanced with fuzzy logic that

quantifies system reliability under uncertainty and supports risk-informed decision-making without relying on historical failure data [32].

4. Future research opportunities

A summary of the suggested areas for future research from the articles reviewed in this section is provided in Table 10. The topics listed above are by no means an exhaustive list of the possible risk-based AIM in the oil and gas industry, but several of them could be applied to other sectors.

More efforts should be made toward developing a risk model that incorporates dynamic models and continuous variables. Most of the risks in the energy sector are gradual and dependent on factors such as corrosion rates, temperature, and pressure. Traditional models, static and discrete, useful for preliminary analyses, frequently oversimplify these dynamic states and can result in missing or less accurate estimates of risk. Moving the risk model in the direction of dynamic processes and continuous data streams will allow for a more realistic representation of complex system behavior. This would improve the capabilities of the risk model to address mechanisms of degradation, the sequence of events in an accident, and dependent risks. Ultimately, this will better inform asset integrity management, operational decision support, and proactive safety design in energy systems.

A significant limitation when developing risk models often continues to be the accessibility of data, leading to dependence on expert judgment and, as a result, subjectivity in the analysis. The use of high-quality data as a future direction for the development of risk models is mentioned as a challenge in some of the studies reviewed above. Although the required data are seldom available at the necessary scale and in the required form, modern developments in data mining techniques could provide interesting possibilities for extracting, processing, and structuring data that could support more robust and objective risk models.

Validation of the model is of fundamental importance for knowing whether the risk model can be adequately implemented for real-world problems; otherwise, it can lead to misleading results. Subsequent studies should always include some form of validation or at least make a case for why this was not possible. Validation enhances models' replicability in the real world, while structural optimization decreases spurious complexity and dependence on expert opinion. The use of simplified structures for models can reduce the time researchers must spend conducting expert elicitation, while still maintaining a model that is useful and informative.

The topics of testing with diverse scenarios, collaborating with industry, and validating on actual systems are all issues associated with the applicability and reproducibility of models. While models are helpful for decision-makers, their actual use can be tricky depending on their level of knowledge around programming and reading models' outputs. Therefore, testing the risk model across diverse scenarios is essential to enhance the robustness, generalizability, and reliability in risk-informed decision making. Furthermore, stakeholder participation is limited to specific sections of the modeling process, rather than being comprehensive, which can result in generic models or oversimplifications. Models included in the analysis should align more closely with practical needs if stakeholders and the industry require the analysis of risks to be conducted using these models.

5. Conclusion

The methodology of risk-based inspection (RBI) is part of the AIM process for assessing equipment in the oil and gas sector. Risk-based AIM is gaining popularity due to its ability to analyze massive equipment and complex processes, distilling information and expertise to prioritize inspection and maintenance. The current study on risk-based AIM has mainly concentrated on risk assessment and the probability of failure. Studies on the consequences of failure assessment, inspection planning, and risk mitigation are limited. Studied risk assessment aimed to establish an existing model that could automate the risk assessment tasks, mitigate the influence of bias from specialist judgment, and ensure the accuracy of risk prediction. In comparison, research on the probability of failure was conducted to determine the likelihood of equipment failure by enhancing the current approach or integrating several failure and condition prediction models. Furthermore, inspection planning and risk mitigation should be studied to improve decision-making by considering execution planning and technique selection to optimize cost without ignoring the risk impact.

The use of risk-based AIM has been reported in process safety, process reliability, process maintenance, structural and mechanical engineering, and computational sciences in over 50 journals, representing the interdisciplinary nature of the field. The increase in publications is consistent annually, demonstrating growing research interest in this area. The analyses of the studies reviewed utilized various quantitative, qualitative, semi-quantitative, probabilistic, deterministic, mixed, dynamic, and traditional risk analysis techniques. The most commonly used analytical techniques were ML, FST, MOO, BN, RSS, MCM, MCDM, and FTA, which show a clear trend towards machine-learning-based techniques as opposed to more traditional approaches. The two types of data considered were: data simulation and actual/historical data, with the latter being more prevalent. The majority of studies analyzed the application at the equipment level, with pipelines being the dominant focus of application.

The main objective of an improved risk-based model is to automate data analysis and reduce the subjective nature of expert assessments. The model facilitates substituting human labour tasks that are very complicated and demand advanced qualifications and high-level thinking abilities. This model enables a streamlined risk-based process, offering data processing capabilities and autonomous data. It minimizes the subjectivity associated with manual assessment, reduces discrepancies in results between assessors, facilitates the analysis and management of high-speed and large volumes of data, and identifies intricate non-linear connections among variables that impact equipment integrity.

Future research on risk-based AIM will increasingly focus on dynamic models, continuous variables with data mining, better data availability, validated and optimized structures, testing across diverse scenarios, and closer collaboration with industry through real-system validation to model applicability and reproducibility.

CRediT authorship contribution statement

Tri Wahono: Writing – original draft, Methodology, Conceptualization. **Agung Purniawan:** Writing – review & editing, Supervision, Methodology. **Imam Mukhlash:** Writing – review & editing, Supervision, Methodology. **Endah R.M. Putri:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Table A1
Distribution of articles according to analysis methods, tools, and objectives.

ID	Ref.	Analysis method*								Analysis tools	Analysis objective
		1	2	3	4	5	6	7	8		
1	[26]	•				•		•		DE	A discrete-event simulation model is used to evaluate the reliability value analysis of complicated repairable systems that link reliability to failure losses.
2	[27]	•				•		•		PCRA	To assess risk by integrating absolute risk (probabilistic) and relative risk (index) models, thereby identifying the most common failure reasons and applying historical incident data.
3	[28]			•		•		•		RBIM	To address personnel limitations, we assess them in detail and determine the most effective maintenance actions and work orders.
4	[29]	•					•	•		BN	A Bayesian probabilistic approach and prediction method are employed to forecast the failure type (rupture or leakage) occurring in steel pipelines in realistic conditions.
5	[30]			•		•		•		GA	To optimize the total cost allocated for the maintenance and repair of the LNG carrier using a genetic algorithm.
6	[31]	•			•			•		FGA	To develop maintenance cost optimization based on fuzzy rule-based and genetic algorithms.
7	[9]	•			•			•		RBI-MOGA	To develop inspection planning based on the integration of RBI and MOGA algorithms.
8	[33]	•			•			•		GA-PSO-BP NN	To develop mitigation strategies by integrating GA, PSO, and BP NN algorithms.
9	[32]			•	•			•		Fuzzy-FTA	The objective of this study is to develop a fuzzy logic–based method integrated with fault tree analysis to address uncertainty and data limitations in reliability assessment, enabling practical risk analysis and informed decision-making.
10	[34]	•			•			•		FE	To develop conditional probability, including the likelihood of a third-party hazard and pipeline fragility.
11	[36]			•	•			•		FL-GA-ER	To develop maintenance optimization based on fuzzy logic, GA, and ER.
12	[35]			•	•			•		Fuzzy BN-BTA	To create a probabilistic safety assessment (PSA) using the BN-bow-tie model to verify critical nodes of accidents by incorporating three metrics of related significance.
13	[37]	•			•			•		FTA-HRA	To integrate fault tree, human reliability, and Monte Carlo simulation to assess LNG spill risks during carrier operations.
14	[41]			•	•			•		FPN	To develop a model of drilling riser risks using FPN theories and fuzzy reasoning by iterating the state matrix to produce risk values.
15	[43]	•			•			•		MM	Develop maintenance planning based on employee scheduling.
16	[38]			•	•			•		FL	To determine the failure probability using past failure data and modification factors.
17	[39]	•				•		•		AL	To evaluate many failure zones with low probability using active metamodel learning and Importance Sampling.
18	[40]	•				•		•		BN	To assess the probability of failure based on fatigue growth and BN.
19	[42]			•	•			•		SPA-LOPA	To propose a quantitative SPA-LOPA method that enhances the accuracy and scientific evaluation of risks and independent protection layers, surpassing the capabilities of traditional semi-quantitative LOPA.
20	[50]	•			•			•		GA-PO	The aim is to develop a decision support system to plan risk-based maintenance through bi-objective optimization.
21	[44]	•					•	•		IFOA-GRNN	To address the challenges in predicting multi-factor complex disasters, we utilise the IFOA-GRNN prediction model in conjunction with an a priori algorithm.
22	[45]			•	•			•		DNVGL	Assess the risk of fish trawling activities in subsea pipelines by calculating the pullover load of the otter board.
23	[51]			•		•		•		NSGA II	To develop decision-making processes for gas supply based on reliability and operational efficiency.
24	[46]	•				•			•	DNN	To predict increases or decreases in risk as system conditions change.
25	[47]	•				•			•	BT-BN	To assess QRA by combining BT and BN diagrams.
26	[48]			•			•	•		MBCT-SR	To assess risks based on pipeline index interval scoring, pipeline failure analysis, and risk level evaluation.
27	[49]	•				•		•		BRANN	To develop a model for transient dispersion studies related to explosions in fixed offshore platforms.
28	[8]			•	•			•		ML	To develop a machine learning–based intelligent system to improve the accuracy, precision, and consistency of risk-based inspection screening assessment in oil and gas facilities.
29	[52]	•				•		•		NSGA II	Develop an optimal plan based on social, economic, and environmental factors.
30	[1]			•	•			•		ML	To optimize the process damage mechanism to support the PoF assessment-based clustering and classification method.
31	[54]	•				•		•		PS-InSAR	Develop ground deformation risk based on seismic data.
32	[55]			•			•	•		Fuzzy DEMATEL-FBLR	To assess pipeline failure analysis based on an index interval scoring method and risk level evaluation.
33	[56]	•				•			•	Fuzzy DEMATEL-BN	To develop a dynamic quantitative risk assessment based on barriers, initiating accident elements, initiating accident events, and risk factors.
34	[57]	•				•		•		CA	To provide a preprocessing tool for grouping samples to simplify the manual risk level assessment or failure prediction procedure.
35	[58]	•				•		•		Fuzzy QRA	To address the uncertainty related to the frequency and severity of consequences.
36	[53]	•				•		•		ML	To develop a quantitative risk analysis model integrating finite element simulation, Latin Hypercube Sampling, and machine learning to assess offshore pipeline failure from falling object collisions under uncertainty.
37	[59]	•				•		•		MC	To develop a risk-based integrity model integrating finite element analysis, probabilistic methods, and cumulative prospect theory to assess offshore pipeline failure from falling objects and optimize protection strategies.
38	[61]			•		•		•		GIS-WOE- BP-GA	To determine the risk zone along the pipeline due to geological hazards, based on factors and geological disasters, using GIS-WOE and the weight of each assessment index by BP-GA.
39	[68]			•		•		•		OWA-AHP	To develop multi-criteria decision-making in gas plant place selection based on risk.
40	[62]			•	•				•	FMEA-VIKOR	To assess risks based on FMEA, utilize the VIKOR extended cloud model to enhance expert judgment.
41	[60]	•			•				•	MKMEF	Probabilistic analysis based on the evolution genetic algorithm (CEGA).

(*continued on next page*)

Table A1 (*continued*)

ID	Ref.	Analysis method*								Analysis tools	Analysis objective
		1	2	3	4	5	6	7	8		
42	[24]	•			•				•	ML	To provide an alternative to traditional physics-based models, we are developing a machine learning-based method for quickly and accurately assessing the risk of failure in corroded oil and gas pipelines. To address the uncertainty about the likelihood and magnitude of impacts.
43	[63]			•		•		•		BT	To develop a cloud-model–based Bow-tie risk assessment method to address uncertainty in probability data and improve prevention and control of fire and explosion accidents in oil depots.
44	[64]	•			•			•		FMEA-VIKOR	To enhance FMEA by integrating Pythagorean fuzzy k-means clustering and PF-VIKOR, thereby addressing limitations in risk assessment, worksheet complexity, and failure mode prioritization.
45	[65]			•	•			•		BN	To develop a decision support framework using Formal Safety Assessment and Bayesian networks to evaluate offshore oil and gas safety risks and identify key hazard factors for preventive decision-making.
46	[66]			•	•				•	SVR-DBN	To develop a hybrid SVR-DBN methodology for dynamic risk assessment of safety-critical equipment on offshore installations, accounting for degradation and human error.
47	[67]			•	•			•		SIMPLE	To develop a methodology for quantifying the economic risk of natural gas transmission pipelines by integrating failure probability estimation, hydraulic analysis, and line pack effects
48	[77]	•				•		•		MDP-DRL	Optimize gas supply reliability using Bayesian networks and deep reinforcement learning.
49	[70]	•				•		•		ML	Essential knowledge must be extracted to improve QRA based on previous risk assessment and ML classifiers, and the implications and frequency of unintended occurrences must be estimated.
50	[25]			•	•			•		API-KM	To assess risk by combining time-independent damage as per the Kent–Muhlbauer method and time-dependent damage as per API-RP 581.
51	[71]	•				•			•	BRANN	To assess the DRA based on non-linear BN features and the non-sequential features of accident escalation.
52	[72]			•		•		•		Fuzzy-TOPSIS	To assess the risk by considering the index of vagueness and randomness based on the triangular fuzzy number, the forward cloud transformation algorithm, and the virtual cloud model.
53	[75]	•				•		•		GCN	To assess risk based on factors affecting accident outcomes and to establish risk assessment indicators.
54	[3]			•	•			•		DWCN-BT-IACO	To develop the probability of risk factors and possible accident-escalation consequences based on integrated algorithms.
55	[74]	•				•			•	DT-ML	To assess risk based on digital twin-machine learning.
56	[69]	•				•		•		GBRT-ANN	To assess RUL using a machine learning approach.
57	[73]	•				•			•	Fuzzy BPA-DSm	To assess risk using a multi-sensor information fusion algorithm.
58	[76]			•		•		•		FC	Enhance the indexing pipeline risk assessment by integrating subtractive clustering fuzzy logic, which accounts for the unpredictability of real-life scenarios.
59	[22]	•				•		•		AI	To assess the PoF based on ML (classification and clustering) and DL algorithms.
60	[81]	•				•		•		RFE-PSO-AB	To assess the susceptibility and vulnerability of pipeline landslides based on a hybrid ML model.
61	[82]	•				•		•		ANN	This research uses an ANN prediction model to evaluate the potential for explosion and fire hazards.
62	[83]	•				•		•		RFECV-CSVR-NSGAIII	To evaluate the risk of excavation using finite elements and machine learning algorithms.
63	[84]	•				•			•	CFC-DRO	To develop the risk of gas pipeline leakage failure based on CRC-DRO.
64	[85]		•			•		•		RSS-ML	To evaluate damage effects and the possibility of catastrophic failures based on RSS and ML.
65	[79]	•				•		•		LWQPSO-ANN	To predict the crater width formed by natural gas pipeline explosions using a nature-inspired optimization algorithm and deep learning.
66	[89]	•				•			•	GA	To develop maintenance optimization based on a meta-heuristic algorithm for MIC.
67	[2]	•				•			•	ML	To develop a text mining–based framework using machine learning, deep learning, and Bayesian networks for automated risk identification and assessment of offshore platforms from hidden danger data.
68	[86]	•				•			•	DBN	To assess the dynamic risk of riderless sound intervention systems that perform P&A operations using DBN.
69	[87]			•		•		•		IV-SFS	To assess the effectiveness of bow tie analysis and IV-SFS in reducing subjectivity in risk assessment.
70	[80]			•		•		•		RCA	This study models the consequences and analyzes the root causes of a real methane pressure vessel explosion in an Iranian gas refinery to support effective risk management and loss reduction in the process industry.
71	[88]			•		•		•		FL	The model presented in this study utilizes a Fuzzy Inference System to assess risk involved in oil and gas pipelines, aiming for higher levels of accuracy and reliability in risk assessment than is achieved through traditional means.
72	[78]			•		•		•		ML	This study proposes a GAN-based framework to generate synthetic pipeline data for enhancing machine learning models in predicting burst failure risk of corroded oil and gas pipelines.
73	[116]			•		•		•		Fuzzy FTA-BN	To assess the probability of failure due to collision based on the hybrid theory.
74	[120]	•				•		•		DF	To develop a pipeline risk assessment model that fuses multiple data sources.
75	[114]	•				•			•	GA-BPNN	To assess the risk of unequal wall thickness of girth weld due to landslides using finite element analysis and deep learning.
76	[118]	•				•		•		AL	To develop active learning strategies for in-line inspection and assess future corrosion risk of offshore pipelines.
77	[119]	•				•			•	BN-DA	To develop a real-time QRA model using a Bayesian network combined with the Dijkstra algorithm.
78	[108]	•				•			•	MCMC	To develop a risk assessment model for dynamic risk assessment by utilizing condition monitoring data.
79	[95]			•	•			•		IFCA	To develop an RBM based on an intuitive fuzzy computational approach.
80	[96]			•		•		•		Fuzzy-MCDM	To develop a risk mitigation strategy that integrates spherical fuzzy sets and MCDM.
81	[23]	•				•		•		ML	To provide inspection planning in line with the implementation of the ML technique.
82	[90]	•				•		•		ML	To predict the failure pressure of corroded gas pipelines based on ML.
83	[91]			•		•			•	DBN-IT2FS	The failure probability based on DBN and fuzzy models was evaluated to minimize the subjectivity and uncertainty of expert appraisal and bias between individual appraisals.
84	[113]	•				•			•	BN	To apply historical data on oil spill accidents and the cause flow chart of accidents in developing a mathematical model for dynamic risk assessment.
85	[92]	•				•			•	MTTF-ML	To predict failure according to a combination of MTTF and ML.

(*continued on next page*)

Table A1 (*continued*)

ID	Ref.	Analysis method*								Analysis tools	Analysis objective
		1	2	3	4	5	6	7	8		
86	[93]	•			•			•		ANN	This study develops a physics- and AI-driven framework for accurately assessing failure risk in corroded gas pipelines to enhance integrity management and safety.
87	[94]	•			•			•		DBN	This work applies uncertain estimates of pipeline corrosion defects through a Bayesian neural network and incorporates the forecast into a Bayesian belief network to perform a complete risk analysis and risk-based integrity management of pipelines.
88	[97]	•			•			•		ML-BN	The present work establishes a data-driven framework for assessing failure risk on oil and gas gathering pipelines by integrating Bayesian networks and LightGBM to support digitalized risk management in Industry 4.0 environments.
89	[99]	•			•			•		DA	This study develops a disaster chain–based framework using expert weighting, interaction matrices, and Dijkstra's algorithm to assess critical risks and evolution paths of oil and gas pipeline leakage in rivers.
90	[103]	•			•			•		DL	This study proposes a predictive framework integrating CFD, deep spatiotemporal modeling, and structural analysis to forecast LNG transfer leaks, assess risks, and guide offshore emergency response.
91	[100]	•			•			•		ML	To enhance risk assessment and safety planning, this study creates and validates a machine learning-based predictive model, using validation through CFD simulations, to calculate the potential impact radius of hydrogen-fueled natural gas pipeline jet fires.
92	[101]	•			•			•		ML	This study develops and deploys a machine learning predictive model, validated through CFD simulations, capable of estimating the impact radius of hydrogen-fueled natural gas pipeline jet fires to support improved risk assessment and safety planning.
93	[104]			•		•		•		Fuzzy-FTA	This study proposes a hybrid AHP–fuzzy method to assess the domino risk of petrochemical pipeline leakage, focusing on fire and explosion accidents to support pipeline risk management and decision-making.
94	[105]	•			•			•		MPF-GCA-FPT-FRA	This study develops a data-driven framework combining MPF simulation, GRA–PCA, FPT prediction, and a corrosion risk matrix to evaluate and optimize inhibition efficiency in oil, gas, and multiphase pipelines.
95	[98]	•			•			•		ML	This study develops machine learning models integrated with fluid hammer analysis to predict the safety status of corroded natural gas pipelines and enhance reliability assessment.
96	[106]			•	•			•		OPA-GraphSAGE	This study proposes a novel UGPN risk assessment model integrating the OPA method and GraphSAGE algorithm to address expert knowledge limitations and incorporate network topology for improved risk management in expanding urban gas pipeline networks.
97	[107]	•			•			•		ML	This study develops a hybrid IGM–ISVM model optimized by IPOA to enhance the accuracy and stability of leakage risk prediction in hydrogen-blended natural gas stations.
98	[102]	•			•			•		DL	This study develops a Physics-Informed Graph Neural Network (Physics_GNN) for accurate real-time simulation and risk assessment of obstructed gas explosions by combining physics-based modeling with deep learning.

*1. Quantitative; 2. Qualitative; 3. Semi-quantitative; 4. Deterministic; 5. Probabilistic; 6. Hybrid deterministic-probabilistic.
7. Traditional risk analysis; 8. Dynamic risk analysis.

Data availability

No data was used for the research described in the article.

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